PolyOCR-Venus: Unified OCR Foundation Models for Text-Centric Visual Intelligence

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Optical Character Recognition (OCR) is evolving from plain-text transcription toward general visual intelligence, requiring models to recognize, localize, and reason over textual information in complex visual environments. However, existing OCR systems often excel at only some tasks and struggle to balance recognition, parsing, and reasoning across scenarios. In this report, we present PolyOCR, a family of unified OCR foundation models of varying scales. PolyOCR combines a shared instruction-following framework with a large-scale data engine that converts heterogeneous visual resources into quality-verified OCR supervision. We introduce Competence-Guided Policy Optimization, which combines verifier-based Group Relative Policy Optimization with on-policy distillation through sample-wise routing based on teacher reliability and the teacher–student competence gap. We also introduce OCRBench v2.1, our revision of OCRBench v2 with manually verified annotation corrections and task-aligned scoring metrics. Extensive experiments across OCRBench v2.1, CC-OCR, in-house KIE Benchmark, OmniDocBench v1.6 and MDPBench demonstrate that PolyOCR achieves state-of-the-art or highly competitive performance.

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Github: https://github.com/inclusionAI/PolyOCR-Venus

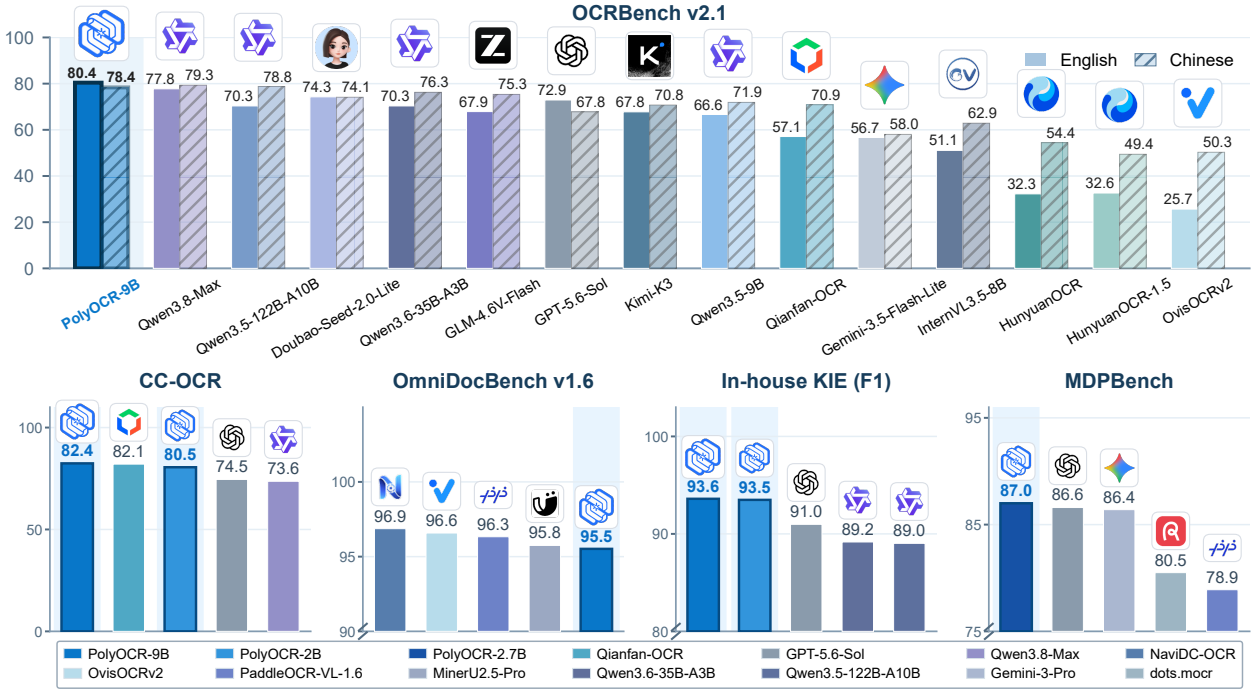


Figure 1 Cross-benchmark comparison. OCRBench v2.1 is shown above; CC-OCR, OmniDocBench v1.6, in-house KIE, and MDPBench are shown below with representative baselines. OmniDocBench uses the PolyOCR-9B layout-guided pipeline, whereas MDPBench uses PolyOCR-2.7B. CC-OCR excludes multilingual OCR; truncated axes are marked.

1 Introduction

Optical Character Recognition (OCR) (Long et al., 2021; Zhang et al., 2024a) has evolved from conventional text transcription toward text-centric visual intelligence (Qu et al., 2023; Chen et al., 2024). Traditional OCR systems mainly focus on converting text-containing images into machine-readable representations through modular pipelines consisting of text detection and recognition (Chaudhuri et al., 2016; Wei et al., 2024). Recent advances in multimodal large language models (MLLMs) have substantially expanded OCR capabilities beyond isolated recognition, enabling scene-text understanding, spatial localization, document structure recovery, table and formula parsing, information extraction, and OCR-oriented reasoning (Fu et al., 2025; Team et al., 2025; Li et al., 2026a; Qu et al., 2026c). As OCR increasingly serves as an interface between visual documents and intelligent systems, future OCR models require not only accurate text recognition but also broad capability coverage and robust generalization across diverse text-centric scenarios.

Despite significant progress, existing OCR systems remain limited in achieving such unified capabilities. Current approaches mainly follow two paradigms. The first consists of specialized OCR systems optimized for specific domains or task families, such as scene text recognition, document parsing, table understanding, and structured information extraction (Team et al., 2025; Cui et al., 2025b; Zhang et al., 2026b; Zhong et al., 2026; Wang et al., 2026a). Although these systems achieve strong performance within their target settings, their task formulations, architectures, supervision schemes, and output formats are often highly specialized, resulting in fragmented capability development and limited knowledge transfer across heterogeneous OCR scenarios. The second paradigm is represented by general-purpose MLLMs, which provide unified interfaces and strong semantic reasoning abilities. However, OCR capabilities in these models are typically acquired implicitly through broad multimodal pretraining rather than systematically optimized across fine-grained text-centric tasks (Achiam et al., 2023; Bai et al., 2025; Xu et al., 2026a). Consequently, they may exhibit uneven performance in precise text recognition, spatial grounding, dense document understanding, and structured content extraction. These limitations reveal a fundamental challenge in balancing capability breadth, task-specific accuracy, and cross-scenario generalization within a single OCR model.

Building a unified OCR foundation model also introduces substantial training challenges. OCR-related tasks differ significantly in visual complexity, output structure, supervision granularity, and optimization objectives. Text recognition requires precise sequence generation, localization requires spatially grounded prediction, structured extraction requires strict format consistency, and OCR-centric reasoning emphasizes semantic understanding. Directly combining these heterogeneous tasks with fixed sampling strategies or static optimization objectives may lead to imbalanced learning dynamics, where dominant tasks overwhelm underrepresented or difficult capabilities. Moreover, positive transfer and task interference can coexist during multi-task learning, while sequential adaptation may cause catastrophic forgetting of previously acquired competencies (Standley et al., 2020; Yu et al., 2020; Zhu et al., 2024; Jiang et al., 2025; Xin et al., 2026b,a). Therefore, an effective OCR foundation model requires not only a unified task formulation but also an adaptive training strategy that coordinates heterogeneous supervision and maintains balanced capability acquisition.

To address these challenges, we present PolyOCR, a family of unified OCR foundation models with 2B and 9B parameter variants built upon Qwen3.5. PolyOCR reformulates heterogeneous OCR and OCR-centric understanding tasks within a shared instruction-following and autoregressive generation framework. Through this unified formulation, diverse outputs, including text transcriptions, spatial coordinates, structured records, document representations, tables, formulas, translations, and open-

form answers, can be generated through a common interface, enabling knowledge sharing across text recognition, text localization, document parsing, information extraction, visual text understanding, and OCR reasoning.

To enable such broad capability coverage, we develop a large-scale OCR data engine that addresses the limited coverage and imbalance of existing public and internal OCR datasets. The data engine converts heterogeneous visual assets into quality-verified supervision through controllable synthesis and model-assisted annotation, with particular emphasis on underrepresented scenarios and complex OCR capabilities that are insufficiently covered by existing resources. We further introduce *Competence-Guided Policy Optimization* (CGPO), a competence-aware post-training framework that integrates verifier-based Group Relative Policy Optimization with on-policy distillation. CGPO dynamically adjusts the distillation weight according to teacher reliability and the evolving teacher–student competence gap, enabling selective and efficient utilization of teacher supervision.

To further extend multilingual document parsing capabilities, we additionally introduce PolyOCR-2.7B, a specialized configuration combining a 1.2B-parameter ViT vision encoder with a 1.5B-parameter Qwen2.5 text decoder. Trained on a curated in-house corpus covering 21 language/script groups, this variant is optimized for robust performance across diverse writing systems and acquisition conditions.

Extensive experiments on OCRBench v2.1, CC-OCR, in-house KIE Benchmark, OmniDocBench v1.6 and MDPBench demonstrate that PolyOCR achieves state-of-the-art or highly competitive performance across diverse OCR scenarios. The model family exhibits strong and balanced capabilities spanning text recognition, text localization, document parsing, information extraction, visual text understanding, and OCR reasoning, establishing the effectiveness of the proposed unified OCR training framework. The main contributions of this work are summarized as follows:

- We introduce PolyOCR, a family of unified OCR foundation models that integrate text recognition, text localization, document parsing, information extraction, visual text understanding, and OCR reasoning within a unified generative framework. The family includes general-purpose variants and a specialized multilingual configuration (PolyOCR-2.7B) optimized for diverse writing systems.
- We develop a large-scale OCR data engine that converts heterogeneous visual resources into quality-controlled multi-task supervision through task-oriented construction, synthesis, and verification. Furthermore, we contribute to the community with refined and comprehensive benchmarks, including OCRBench v2.1, CC-OCR, in-house KIE Benchmark, OmniDocBench v1.6 and MDPBench, to facilitate rigorous evaluation across general, multilingual, and industrial OCR scenarios.
- We propose *Competence-Guided Policy Optimization* (CGPO), a competence-aware post-training framework that dynamically combines verifier-based policy optimization and on-policy distillation through sample-wise supervision routing, effectively balancing heterogeneous task learning and preventing catastrophic forgetting.

2 Related Work

2.1 Specialized OCR Systems

Optical Character Recognition has traditionally been developed through task-specific formulations, including text detection, recognition, text spotting, handwriting recognition, multilingual OCR, and mathematical expression recognition (Long et al., 2021; Zhang et al., 2024a). Early OCR systems commonly adopted modular pipelines, where text localization, visual recognition, and post-processing were optimized independently. Representative approaches such as EAST (Zhou et al., 2017) and CRNN (Shi et al., 2016) established effective paradigms for scene-text detection and recognition, while recent Transformer-based models, including TrOCR (Li et al., 2023) and PARSeq (Bautista and Atienza, 2022), further improved recognition accuracy through large-scale pretraining and sequence modeling.

Beyond text recognition, specialized OCR models have been developed for diverse scenarios (Qu et al., 2025a, 2024a), including handwriting recognition, multilingual text understanding, formula recognition, text spotting, and structured information extraction. These systems typically employ task-specific architectures, supervision formats, decoding mechanisms, and evaluation protocols tailored to individual objectives. Although such specialization enables strong performance within specific domains, it also leads to fragmented capability development, where knowledge learned from one OCR task is difficult to transfer to others with different visual characteristics, output structures, and reasoning requirements.

The increasing diversity of OCR applications calls for a unified modeling paradigm that learns heterogeneous OCR capabilities within a shared framework (Qu et al., 2024b, 2026b).

2.2 Document Intelligence Models

Document intelligence extends OCR from isolated text recognition toward page-level understanding, including layout analysis, reading-order modeling, structured reconstruction, and semantic information extraction (Qu et al., 2026d). Early approaches such as LayoutLM (Xu et al., 2020) and LayoutLMv3 (Huang et al., 2022) jointly modeled textual, visual, and spatial information, enabling effective document-level representation learning.

Another line of research explores OCR-free document understanding, where document images are directly converted into structured outputs without explicit OCR modules. Donut (Kim et al., 2022) introduced an end-to-end document understanding framework based on image-to-sequence generation, while Nougat (Blecher et al., 2024) focused on converting scientific documents into structured markup representations. Recent document parsing systems further improve full-page understanding through structure-aware modeling and large-scale multimodal training, including MonkeyOCR (Li et al., 2025), Dolphin (Feng et al., 2025, 2026), MinerU2.5 (Niu et al., 2026; Wang et al., 2026a), and PaddleOCR-VL (Cui et al., 2025a; Zhang et al., 2026b).

Despite significant progress in document-level understanding, existing document intelligence models primarily optimize for page-centric scenarios. Their capabilities are often concentrated on layout analysis, structural reconstruction, and document semantic extraction, while broader OCR intelligence, including natural-scene text perception, multilingual recognition, region-conditioned understanding, and OCR-oriented reasoning across heterogeneous visual environments, remains insufficiently unified.

2.3 OCR-oriented Vision-Language Foundation Models

The emergence of multimodal large language models provides a unified generative paradigm for visual perception, language understanding, spatial reasoning, and structured output generation. General-purpose vision-language models, including Gemini (Comanici et al., 2025), Qwen3-VL (Bai et al., 2025), InternVL3.5 (Wang et al., 2025b), and Ovis2.5 (Lu et al., 2025), demonstrate strong capabilities across a wide range of visual understanding tasks. By adopting instruction following as a common interface, these models can flexibly handle image understanding, visual question answering, grounding, and document-related tasks within a unified framework.

Despite their broad multimodal capabilities, OCR competence in general-purpose vision-language models is typically acquired as an auxiliary capability from large-scale multimodal pretraining rather than through systematic optimization over diverse text-centric tasks. As a result, these models often exhibit uneven performance across fine-grained character recognition, dense text perception, precise localization, structured information extraction, and document-level reasoning. Comprehensive OCR evaluations, including OCRBench (Liu et al., 2024) and OCRBench v2 (Fu et al., 2025), reveal substantial performance variations among different OCR capability dimensions, indicating that a unified multimodal interface alone is insufficient to establish balanced OCR intelligence.

To address this limitation, recent studies have explored OCR-oriented vision-language models that explicitly optimize text-centric visual understanding. TextMonkey (Liu et al., 2026) improves high-resolution text perception and grounding through OCR-oriented visual modeling. GOT-OCR2.0 (Wei et al., 2024) extends OCR beyond plain-text transcription by supporting recognition, formulas, tables, charts, and region-conditioned understanding within a unified generative framework. HunyuanOCR (Team et al., 2025; Li et al., 2026a) integrates text spotting, document parsing, information extraction, translation, and OCR-oriented question answering into a single OCR model. OCRVerse (Zhong et al., 2026) and Qianfan-OCR (Dong et al., 2026) further expand OCR vision-language modeling toward broader task coverage and structured visual-text understanding.

These OCR-oriented vision-language models represent an important transition from isolated OCR modules toward unified generative OCR systems. Nevertheless, existing approaches still differ in their primary optimization objectives, with some emphasizing recognition and grounding, some focusing on document parsing, and others targeting selected extraction or reasoning tasks. A comprehensive OCR foundation model requires further integration of heterogeneous OCR capabilities, balanced supervision across diverse tasks, and adaptive optimization strategies that can maintain consistent performance from low-level text perception to high-level visual-text reasoning.

2.4 OCR Data Construction and Capability-aware Post-training

The development of OCR foundation models relies on large-scale and diverse supervision. Existing approaches typically combine manually annotated datasets with web-collected documents, scanned books, natural scene images, synthetic text images, rendered tables and formulas, automatically parsed documents, and model-generated instruction data (Cui et al., 2025b; Wang et al., 2026a; Dong et al., 2026; Jiang et al., 2026a,b). Synthetic generation and model-assisted annotation have become important for expanding coverage of long-tail scenarios, complex layouts, and structured output formats. However, constructing large-scale OCR supervision remains challenging due to annotation cost, heterogeneous task requirements, and the difficulty of maintaining consistent quality across diverse sources.

Recent OCR-oriented models commonly adopt multi-task supervised fine-tuning to align pretrained

multimodal models with heterogeneous OCR tasks and output formats (Li et al.). Knowledge distillation and reinforcement learning have further been explored to improve recognition accuracy, structural validity, and instruction-following ability (Team et al., 2025; Li et al., 2026a; Zhang et al., 2026b; Zhong et al., 2026). Nevertheless, OCR tasks differ substantially in data availability, visual complexity, annotation reliability, and optimization difficulty. Fixed data sampling strategies or static loss weighting may lead to imbalanced capability acquisition, where frequently optimized tasks dominate training while challenging or underrepresented capabilities remain insufficiently learned.

Recent OCR training pipelines provide more direct precedents for outcome optimization and distillation. olmOCR 2 uses the pass rate of verifiable unit tests as a GRPO reward, assessing text fidelity, reading order, table relations, and formula rendering (Poznanski et al., 2025). Infinity-Parser2 performs joint reinforcement learning over eight document and multimodal tasks with task-specific rewards for textual, spatial, and structural accuracy (Huang et al., 2026). OvisOCR2 transfers an RL-enhanced 4B teacher to a 0.8B student through on-policy distillation with a reverse-KL objective evaluated on the student's top-k support (Lu et al., 2026).

Beyond OCR, generalized knowledge distillation (GKD) already combines reinforcement learning with on-policy teacher supervision (Agarwal et al., 2024). More recent methods make this supervision conditional on observed outcomes: SCOPE routes verified student trajectories to perplexity-weighted self-reinforcement or distillation (Zheng et al., 2026), whereas Reward-Gated OPD selects trajectories according to agreement between verifier feedback and the teacher–student likelihood gap (Akhondzadeh et al., 2026). CGPO instead estimates competence from teacher and student generations scored against a shared reference, calibrates teacher quality and probability of superiority on held-out generations, and uses the resulting sample-level weight to modulate OPD alongside an always-active GRPO objective.

Adaptive optimization methods have been investigated in broader multi-task and model training scenarios (Qu et al., 2026a, 2025b). TwSG (Jiang et al., 2026c) proposes the TL-GRPO scheme to optimize model performance on text-intensive understanding tasks such as ChartQA. GradNorm (Chen et al., 2018) balances task losses according to learning dynamics, PCGrad (Yu et al., 2020) mitigates gradient conflicts, DoReMi (Xie et al., 2023) optimizes data mixtures through distribution reweighting, and Skill-it! (Chen et al., 2023) adapts training schedules according to skill dependencies. However, these approaches mainly focus on task balancing, data allocation, or curriculum design, without explicitly modeling the evolving competence relationship between teacher and student models during OCR-specific post-training (Yu et al., 2024). Building unified OCR foundation models therefore requires both comprehensive supervision and capability-aware optimization strategies that can adaptively allocate learning signals according to model competence.

3 Data Construction

A unified OCR foundation model requires large-scale and diverse supervision that covers the full spectrum of text-centric visual intelligence capabilities, ranging from fundamental text recognition and spatial localization to structured content recovery, information extraction, multilingual transformation, and OCR-oriented reasoning. However, existing OCR datasets are typically constructed for individual tasks, visual domains, or languages, resulting in fragmented capability coverage and imbalanced supervision across different OCR scenarios.

To address this challenge, we develop a comprehensive data construction framework consisting of

two complementary components: full-spectrum OCR data collection and targeted capability-driven data construction. First, we curate large-scale public datasets and in-house resources to establish broad coverage across OCR tasks, visual environments, languages, and annotation formats. Second, for capabilities that remain underrepresented due to limited public resources or expensive annotation requirements, we develop a targeted data-construction engine that generates controllable synthetic and model-assisted supervision. Together, these two components provide a unified and scalable data foundation for training OCR foundation models.

3.1 Full-Spectrum OCR Data Collection

Inspired by Venus-DeFakerOne (Team, 2026), to establish broad capability coverage for unified OCR training, we curate OCR data from diverse visual domains and organize them according to six fundamental capability dimensions: text recognition, text localization, document parsing, information extraction, visual text understanding, and OCR reasoning. We use these six dimensions as a benchmark-independent organization of the training corpus. Figure 2 illustrates representative tasks within these dimensions, and Table 1 summarizes their organization in the training corpus.

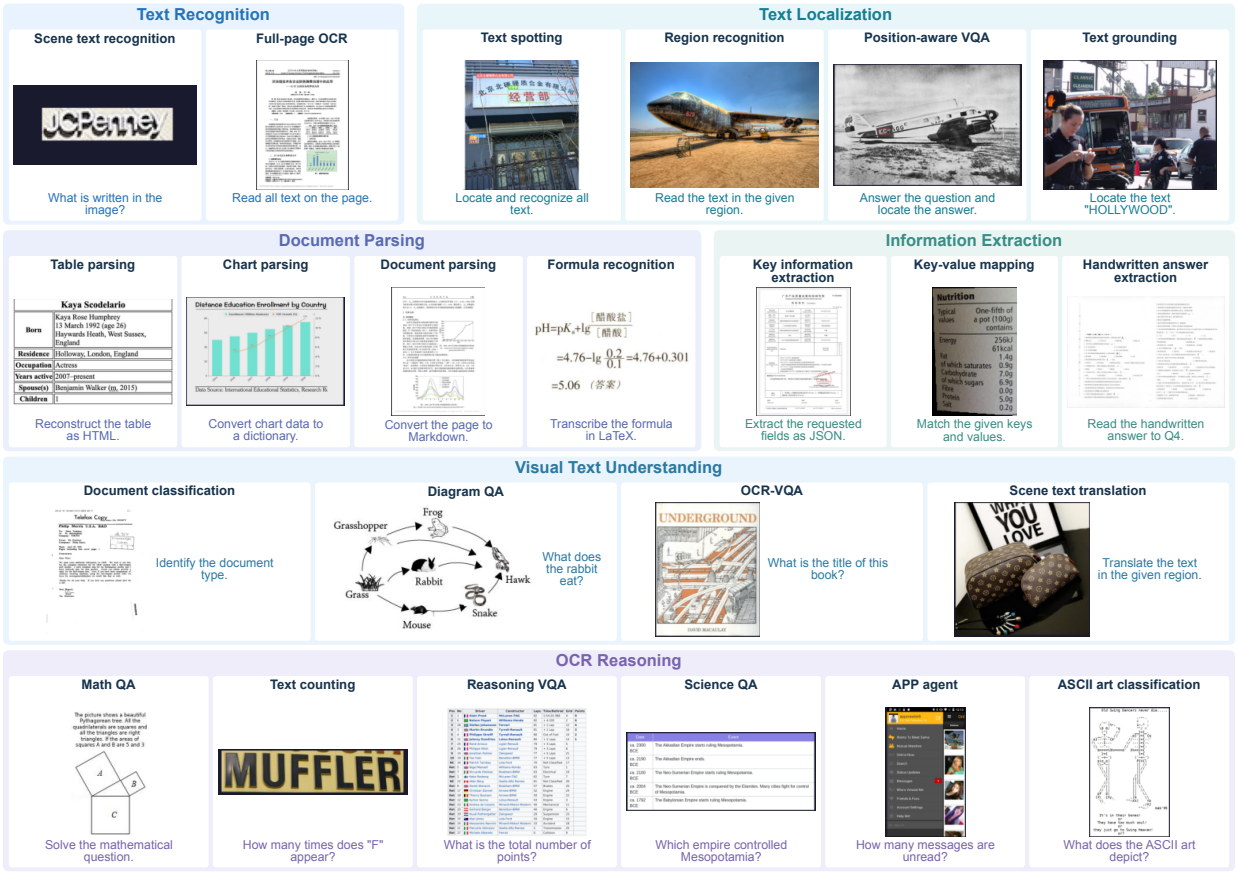


Figure 2 Representative tasks across the six OCR capability dimensions. Each card pairs an example image with a shortened instruction. The gallery combines task illustrations from Table 16 with selected examples from OCRBench v2 (Fu et al., 2025); the displayed examples illustrate task definitions rather than the composition of the training corpus.

The examples illustrate how a visual domain can support different tasks: a document may require transcription, structure recovery, field extraction, or question answering. Organizing data by the

requested operation distinguishes these supervision requirements while retaining coverage across scenes, documents, charts, and application interfaces.

Text recognition. This category covers fundamental text perception capabilities across different visual conditions and granularity levels, including printed text recognition, scene text recognition, handwriting recognition, multilingual recognition, curved text recognition, and dense document transcription. The collected supervision spans character-, word-, line-, paragraph-, and page-level recognition tasks (Ying et al., 2026; Karatzas et al., 2015; Ch'ng and Chan, 2017; Marti and Bunke, 2002; Liu et al., 2019; Jiang et al., 2023; Yuan et al., 2022).

Text localization. This category focuses on spatially grounded text understanding, including text detection, text grounding, referring text localization, region-conditioned recognition, and text spotting. The corresponding annotations provide explicit spatial supervision represented by bounding boxes, quadrilaterals, or polygons, enabling joint learning of textual content and spatial information (Karatzas et al., 2015; Ch'ng and Chan, 2017; Veit et al., 2016; Long et al., 2022).

Document parsing. This category includes document layout analysis, reading-order recovery, full-page document parsing, table reconstruction, chart parsing, and mathematical expression recognition. These tasks require models to recover hierarchical document structures beyond plain-text transcription, including both printed and handwritten mathematical content (Zhong et al., 2019, 2020; Smock et al., 2022; Cheng et al., 2023; Gervais et al., 2024).

Information extraction. This category covers structured information understanding from visually rich documents, including entity extraction, key information extraction, key–value association, form understanding, and semantic relation recovery from receipts, forms, certificates, and other structured document images (Jaume et al., 2019; Huang et al., 2019; Xu et al., 2022; Kuang et al., 2023).

Visual text understanding. This category extends OCR beyond perception toward higher-level visual-text interaction, including scene and document question answering, document classification, chart and table understanding, diagram interpretation, and visual text translation. These tasks require models to interpret textual information in the context of surrounding visual structures and perform task-oriented transformations (Singh et al., 2019; Biten et al., 2019; Mathew et al., 2021; Masry et al., 2022; Mishra et al., 2019; Liang et al., 2024).

OCR reasoning. This category targets reasoning capabilities grounded in visual text, including character and text-instance counting, arithmetic operations over visible values, spatial relationship reasoning, scientific question answering, and multi-step reasoning over scenes, documents, tables, charts, diagrams, and application interfaces (Masry et al., 2022; Lu et al., 2022, 2024; Zhang et al., 2024b).

Although the collected corpus provides broad coverage across OCR tasks and visual scenarios, naturally available datasets remain highly imbalanced. In particular, capabilities such as spatially grounded bilingual translation, template-based key information extraction, Chinese document and table parsing, scene-text localization, and OCR-centric reasoning lack sufficient high-quality supervision due to limited public resources or expensive annotation processes. Therefore, we further develop a targeted data-construction engine to generate controllable synthetic and model-assisted training data for these underrepresented capabilities.

Table 1 Capability organization of the full-spectrum OCR training corpus.

Capability group	Representative tasks
Text Recognition	Scene text, handwriting, multilingual OCR, full-page transcription
Text Localization	Detection, grounding, referring, region recognition, text spotting
Document Parsing	Document, layout, table, formula, chart parsing
Information Extraction	Entity extraction, KIE, key–value association, form understanding
Visual Text Understanding	Document VQA, chart/table understanding, classification, translation
OCR Reasoning	Counting, numerical calculation, spatial reasoning, multi-step inference

3.2 Targeted Data-Construction Engine

Although full-spectrum data collection provides broad OCR coverage, naturally available datasets remain insufficient for several capabilities due to limited annotations and high labeling costs. To address these gaps, we develop a targeted data-construction engine that generates capability-specific synthetic and model-assisted supervision. Each pipeline is designed to preserve the semantic requirements, spatial constraints, and output formats of its target OCR task.

As illustrated in Figure 3, the engine decouples raw visual resources from task-specific supervision generation. Diverse sources, including street images, document templates, annual reports, books, PDFs, tables, and verified OCR annotations, are transformed into supervision for scene-text understanding, bilingual translation, key information extraction, document parsing, table understanding, and OCR-centric reasoning. The engine is organized into three complementary synthesis pipelines: scene-text and translation synthesis, structured KIE, document, and table synthesis, and reasoning-oriented annotation synthesis. Each pipeline incorporates dedicated generation and verification procedures to ensure visual-text consistency, structural validity, and task-specific supervision quality.

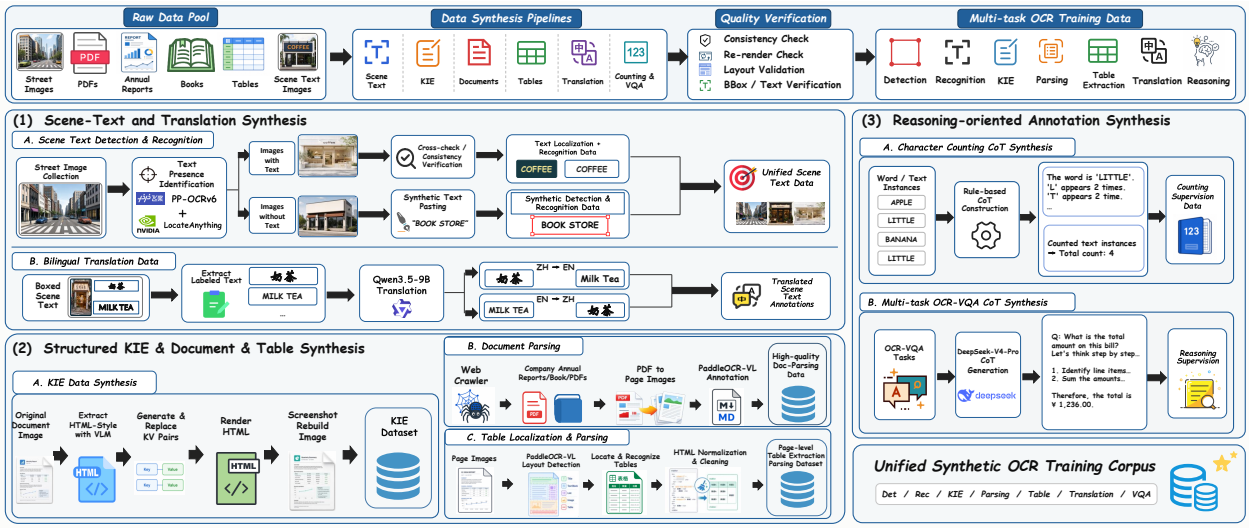


Figure 3 Overview of the targeted data-construction engine. Heterogeneous raw assets are transformed into task-specific supervision for scene-text detection and recognition, bilingual translation, key information extraction, document parsing, Chinese table localization and parsing, and OCR-centric reasoning.

(1) Scene-Text and Translation Synthesis

A. Scene Text Detection & Recognition. Rendering text into natural backgrounds provides paired localization and recognition supervision, as demonstrated by SynthText (Gupta et al., 2016). We

construct scene-text supervision from both real-world images and controllable synthesis. For text-containing natural images, PP-OCRv6 (Zhang et al., 2026a) and LocateAnything (Wang et al., 2026b) are employed as independent annotators for text localization and recognition. Their predictions are cross-validated through spatial overlap and transcription consistency, and only mutually verified instances are retained as localization, recognition, and spotting supervision. Samples exhibiting substantial disagreement are identified as hard cases and further annotated by human annotators before being incorporated into the training corpus. For images without visible text, a synthetic branch renders text onto realistic background regions while controlling language, font, character size, orientation, color, contrast, perspective distortion, blur, and compression artifacts. The rendered transcriptions and spatial annotations are further verified for visual-text consistency before being added to the training set. This dual-source construction strategy provides both naturally occurring scene-text diversity and controllable localization-recognition supervision.

B. Bilingual Translation Data. We construct spatially grounded translation supervision from scene images with verified text regions and transcriptions. For each text instance, Qwen3.5-9B generates both Chinese–English and English–Chinese translations. The generated translations are filtered using language consistency, numerical and symbol preservation, named-entity consistency, repetition detection, and length constraints. Each sample contains the original image, a region-aware translation instruction, and the corresponding translation. This construction enables the model to jointly learn text localization, recognition, and multilingual transformation while maintaining the spatial grounding of the source text.

(2) Structured KIE & Document & Table Synthesis

A. KIE Data Synthesis. Recent KIE synthesis frameworks such as SAYRE use document exemplars and editable HTML representations to generate new images with aligned field annotations (Xu et al., 2026b). We develop a template-preserving synthesis pipeline for structured documents, including forms, receipts, certificates, labels, and other field-oriented images. A vision-language model first reconstructs each document into an editable HTML/CSS representation. A language model then replaces the original key–value contents while preserving field semantics and layout constraints. The modified templates are subsequently rendered into new document images with naturally aligned annotations. Generated samples are filtered for text overflow, abnormal line wrapping, element overlap, missing fonts, field inconsistency, and severe visual distortion. This pipeline expands the semantic and visual diversity of KIE supervision while preserving accurate key–value alignment.

B. Document Parsing. To enrich document-level supervision, we collect a diverse corpus of PDF documents, including academic papers, annual reports, research reports, books, and documents containing handwritten content, and render their pages as images. To promote page diversity, we first extract page features using a visual encoder and apply k-means clustering to group pages with similar visual characteristics. In parallel, we use PP-DocLayoutV3 to detect layout elements and compute a page complexity score based on their counts and predefined category weights, assigning higher weights to complex, less frequent element types, such as figures and charts, and lower weights to common, simpler elements, such as regular text. Within each cluster, we stratify pages by their complexity scores and sample uniformly across strata, ensuring coverage of diverse visual characteristics and complexity levels.We then independently parse the sampled pages using PaddleOCR-VL and MinerU2.5-Pro to produce document annotations in reading order (Cui et al., 2025b,a; Wang et al., 2026a). We compare the outputs of the two models and use their similarity as a proxy for annotation quality. Annotations for pages with high inter-model agreement are retained as high-quality candidates. For pages with low agreement, we use Qwen-3.5 to identify and correct

omissions, recognition errors, and structural discrepancies, taking both parsing outputs and the original page image as input. The resulting annotations are consolidated into Markdown and plain-text formats. Markdown annotations preserve document hierarchy and content elements, including paragraphs, headings, notes, lists, tables, and formulas, while plain-text annotations support full-page transcription tasks. All annotations undergo rule-based cleaning and normalization, including LaTeX standardization for formulas, HTML normalization for tables, removal of unsupported tags and placeholders, elimination of malformed structures, and deduplication.

C. Table Localization & Parsing. Existing datasets for table understanding primarily focus on table structure recognition, offering limited support for joint table localization and parsing within full-page documents. To address this limitation, we construct a full-page table understanding dataset covering diverse visual styles and levels of structural complexity. We first detect tables in document pages and classify them as fully bordered, borderless, or partially bordered using a table classification model. Each table is then independently parsed by PaddleOCR-VL and MinerU2.5-Pro (Cui et al., 2025b,a; Wang et al., 2026a), and their outputs are normalized into HTML. We compute the TEDS score between the corresponding outputs to assess inter-model consistency as a reference for annotation quality. Annotations with high consistency are accepted as candidates, while those with lower consistency are refined by Qwen-3.5 using the original page image and both models' outputs. To preserve annotation completeness, we retain a page only after all its tables have been annotated and passed the quality checks, discarding pages with unresolved omissions or severe annotation errors. From the resulting HTML annotations, we extract row and column counts, the number and proportion of merged cells, and their row and column spans to characterize table size and structural complexity. We use these attributes together with the visual style categories for stratified sampling, followed by manual spot checks to promote data diversity and annotation quality. Each sample comprises a full-page document image, table bounding boxes, and HTML annotations, supporting joint table localization and structure parsing.

(3) Reasoning-oriented Annotation Synthesis

A. Character Counting CoT Synthesis. To provide reliable supervision for OCR-related quantitative reasoning, we construct counting tasks from quality-verified text recognition and localization annotations using deterministic templates. Text annotations are normalized with respect to character encoding, whitespace, punctuation, and full-width or half-width symbols while preserving character case and visual semantics. Based on these annotations, we automatically generate diverse counting instructions and reasoning traces, including character-level counting and text-instance counting. The generated samples inherit exact answers from verified OCR annotations, providing consistent supervision for OCR-centric numerical reasoning without introducing noise from model-generated labels.

B. Multi-task OCR-VQA CoT Synthesis. For complex reasoning tasks that cannot be efficiently generated through deterministic rules, we employ strong reasoning models, represented by DeepSeek-V4-Pro (Xu et al., 2026a), to construct candidate questions, explanations, and answers based on verified OCR and structured annotations. The generated examples cover textual content understanding, document field reasoning, key–value relationships, table interpretation, spatial reasoning, and other OCR-centric capabilities. The source annotations are used only for generation guidance and verification. Candidate samples are validated through task-specific consistency checks, and examples containing incorrect answers, unsupported reasoning steps, or invalid output formats are removed.

3.3 Cross-Pipeline Quality Control

Across the construction pipelines, quality control uses three complementary forms of verification. First, agreement-based validation compares independent annotations when multiple recognizers or parsers are available, retaining mutually consistent predictions and sending selected scene-text disagreements for human adjudication. Second, structure-aware validation checks whether generated coordinates, HTML, Markdown, formulas, and key–value records remain well formed and consistent with the rendered image. This includes detecting element overlap, text overflow, malformed table structures, missing fields, unsupported tags, and content loss. Third, source-grounded validation checks whether generated translations, questions, reasoning traces, and answers preserve the textual or structured evidence available in the source annotations. Samples that fail the corresponding checks are removed rather than repaired through unconstrained generation.

These validation signals also expose difficult examples. In particular, cross-model disagreements that are resolved through human annotation provide quality-verified hard scene-text cases, while valid samples with challenging structured outputs are retained only after passing task-specific format and consistency checks. During training, such hard examples are mixed with broader coverage data without being allowed to dominate the task distribution. After capability balancing and sampling, the resulting corpus delivers approximately 60 million high-quality OCR training instances, achieving broad coverage across diverse scenarios, tasks, and supervision formats to support model training.

4 Training

4.1 Training Pipeline

Training a unified OCR foundation model is challenging due to the highly heterogeneous nature of OCR tasks, where different capabilities exhibit distinct learning difficulties, optimization objectives, and competence boundaries. A model may achieve strong performance on common recognition patterns while still struggling with complex document structures, long-tail scenarios, or reasoning-intensive OCR tasks.

To address this challenge, we develop a two-stage training framework consisting of capability acquisition and competence-guided refinement. The first stage adopts supervised fine-tuning (SFT) to establish broad OCR capabilities across heterogeneous tasks, including text recognition and localization, document parsing, table understanding, key information extraction, multilingual transformation, and OCR-centric visual question answering. The second stage introduces Competence-Guided Policy Optimization (CGPO), which combines verifier-based Group Relative Policy Optimization (GRPO) with adaptive on-policy distillation (OPD) to dynamically allocate optimization signals according to the competence relationship between the teacher and student models.

Both model variants use the same SFT and CGPO training settings. The CGPO stage uses a learning rate of 10^{-6}, with Qwen3.5-122B-A10B serving as the teacher for offline competence estimation and token-level OPD.

4.2 Supervised Fine-Tuning

The SFT corpus contains task families that differ substantially in data volume, output structure, and learning difficulty. Sampling them in their raw proportions would overrepresent the largest sources and provide limited control over the contribution of each capability. We therefore partition the corpus into four groups and assign their sampling quotas explicitly. Text Perception (TP) covers text

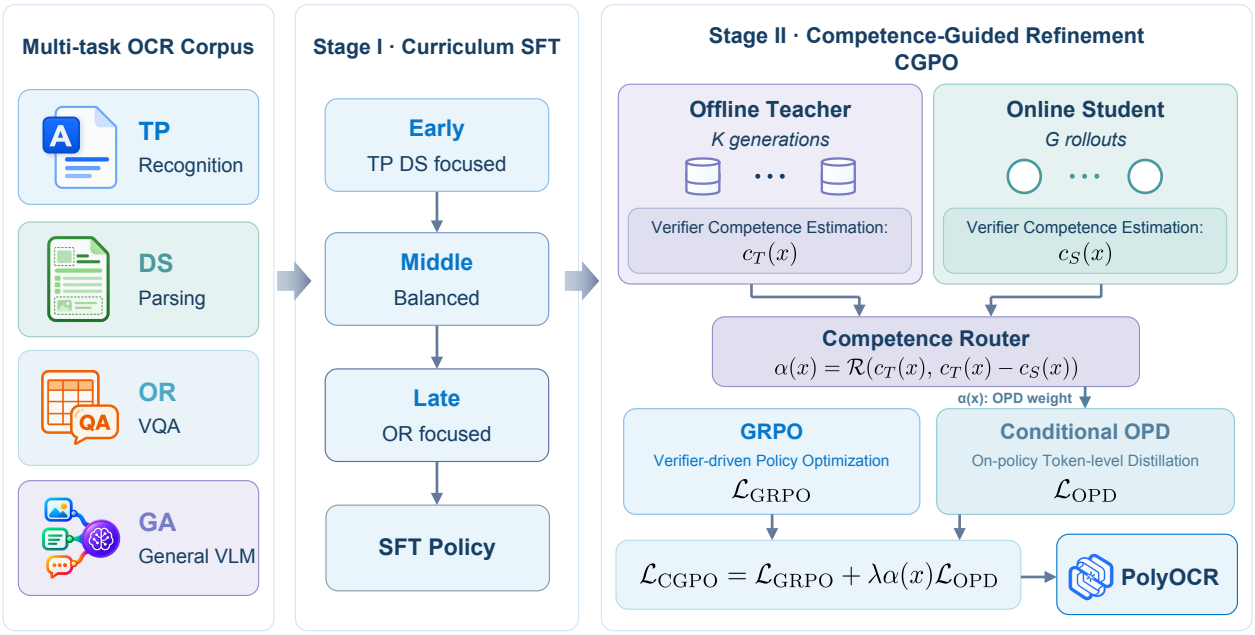


Figure 4 Overview of the PolyOCR training framework. Stage I applies a three-phase curriculum SFT schedule over Text Perception (TP), Document Structure Understanding (DS), OCR Understanding and Reasoning (OR), and General Multimodal Alignment (GA) data. Stage II performs CGPO by combining verifier-driven GRPO with conditional OPD. Here, $\mathcal{R}$ denotes the calibrated competence-routing map defined in Sec. 4.3, which maps the cached teacher competence $c_T(x)$ and teacher–student competence gap $c_T(x) - c_S(x)$ to the OPD weight $\alpha(x)$.

recognition, grounding, position-aware VQA, and text spotting. Document Structure Understanding (DS) includes key-information extraction and mapping, handwritten answer extraction, document, chart, and table parsing, and formula recognition. OCR Understanding and Reasoning (OR) comprises classification, translation, counting, mathematical question answering, and other OCR-grounded reasoning tasks. General Multimodal Alignment (GA) retains broad image understanding and instruction-following data.

Across the approximately 60 million SFT instances, TP, DS, OR, and GA account for 32.5%, 30.0%, 22.5%, and 15.0%, respectively. This allocation gives most of the optimization budget to text perception and document structure, while retaining dedicated coverage for OCR-grounded semantics, reasoning, and general multimodal behavior.

Table 2 Three-phase sampling schedule for SFT. Percentages denote the within-phase proportions of the four data groups.

Phase	Instances	TP	DS	OR	GA
Early	20M	40.0%	35.0%	10.0%	15.0%
Middle	20M	32.5%	30.0%	22.5%	15.0%
Late	20M	25.0%	25.0%	35.0%	15.0%
Overall	60M	32.5%	30.0%	22.5%	15.0%

We apply a three-phase quota schedule, with each phase containing 20 million instances. The early phase places greater weight on TP and DS. The OR share then increases from 10.0% to 35.0% as training progresses, whereas GA remains fixed at 15.0%. Each group is sampled without replacement according to its phase-specific quota, and the selected instances are shuffled within

each phase. Consequently, the complete schedule traverses the full SFT corpus while preserving the overall mixture shown in Table 2. The resulting checkpoint initializes the subsequent CGPO stage.

4.3 Competence-Guided Policy Optimization

The SFT stage establishes broad OCR competence, but its imitation objective does not directly optimize verifier-defined task outcomes. We therefore use outcome-supervised Group Relative Policy Optimization (GRPO), which assigns each response a scalar reward and derives token advantages from its relative reward within a rollout group (Shao et al., 2024). Our ablation in Table 9 reports a CC-OCR score of 81.72 for GRPO and 82.18 after adding fixed-weight OPD, supporting the use of complementary supervision.

Consider an illustrative long-text OCR case: a financial document contains 1,000 characters after normalization, including the amount `18,650`. Suppose the eight student rollouts preserve the surrounding text but transcribe this amount as `18,850` or `18,350`. Each output contains one character substitution, so normalized edit similarity assigns all eight responses the same reward, $1 - 1/1000 = 0.999$. The amount is incorrect despite the high page-level score. With $A_i = (r_i - \bar{r})/(\mathrm{std}(r) + \epsilon)$, all group-relative advantages are zero, and this group contributes no reward-driven policy gradient to correct the digit. This equal-reward degeneracy is also discussed in DAPO (Yu et al., 2025). Even when rewards differ, the response-level advantage does not identify the erroneous position.

On-policy distillation (OPD) supplies dense token-level supervision by evaluating the teacher on student-generated prefixes (Agarwal et al., 2024). In the example, a teacher distribution that favors the correct continuation when the amount is generated provides information that the tied outcome rewards cannot supply. However, this supervision inherits the teacher's errors: if the teacher favors `18,850`, distillation can reinforce the same incorrect digit. If the student later recognizes `18,650` correctly while the teacher still favors the wrong continuation, a fixed OPD weight can oppose the correction encouraged by the verifier. Teacher fluency and a high page-level score therefore do not establish the usefulness of its local guidance.

We propose *Competence-Guided Policy Optimization* (CGPO) to retain GRPO as the optimization backbone and condition the OPD weight on both teacher quality and the teacher's advantage over the current student. Reliable teacher advantages warrant stronger distillation; unreliable guidance or a diminishing advantage warrants less. This sample-wise allocation preserves verifier-guided exploration as student competence changes.

For each training sample x from task family τ, let $y^\star$ denote its quality-verified reference. A task-specific verifier assigns an outcome reward $R_\tau(x, y, y^\star) \in [0, 1]$. The verifier is instantiated according to the semantic structure of each task, using normalized edit similarity for text recognition, intersection-over-union (IoU) for visual grounding, character detection matching (CDM) for formula recognition (Wang et al., 2025a), tree-edit-distance-based similarity (TEDS) for table parsing (Zhong et al., 2020), field-level F1 for key information extraction, and an LLM-based correctness judge for open-ended OCR-centric VQA. Task-specific rewards also underpin multi-task OCR reinforcement learning in Infinity-Parser2 (Huang et al., 2026), while criterion-based language-model assessment has been studied in G-Eval (Liu et al., 2023). Our task-specific metrics are mapped to a common reward space and supplied to GRPO for outcome-based policy optimization. The same verified reference and verifier are used to score teacher and student generations.

Competence estimation. Before CGPO optimization, the teacher is evaluated on each training sample by sampling K responses:

$$r_{T,k}(x) = R_\tau\left(x, y_T^{(k)}, y^\star\right), \qquad c_T(x) = \frac{1}{K}\sum_{k=1}^{K} r_{T,k}(x), \tag{1}$$

where $c_T(x)$ provides a verifier-based estimate of the teacher's competence on sample x. This quantity is computed before policy optimization and cached throughout training.

During CGPO optimization, the current student policy generates G responses for the same sample, yielding an online estimate of student competence:

$$r_{S,i}(x) = R_\tau\left(x, y_S^{(i)}, y^\star\right), \qquad c_S(x) = \frac{1}{G}\sum_{i=1}^{G} r_{S,i}(x). \tag{2}$$

The teacher–student competence gap is then defined as:

$$\Delta(x) = c_T(x) - c_S(x). \tag{3}$$

A positive $\Delta(x)$ indicates that the teacher remains ahead of the student on the current sample, whereas a small or negative value indicates that the student has approached or exceeded the teacher's competence. Importantly, a positive mean gap alone is insufficient to establish useful supervision: a low-quality teacher may still outperform a weaker student, and a gap estimated from finitely many rollouts may not reproduce under an independent draw.

Offline routing calibration. We determine the routing map on a fixed calibration split drawn from the quality-verified training pool using task- and difficulty-stratified sampling. For each calibration sample, teacher and SFT-student generations are divided into independent sets A_T, B_T and A_S, B_S. The A sets emulate the finite-rollout statistics available to the router,

$$\widehat{c}_T^{A}(x) = \frac{1}{|A_T|}\sum_{k\in A_T} r_{T,k}(x), \qquad \widehat{c}_S^{A}(x) = \frac{1}{|A_S|}\sum_{i\in A_S} r_{S,i}(x), \qquad \widehat{\Delta}^{A}(x) = \widehat{c}_T^{A}(x) - \widehat{c}_S^{A}(x), \tag{4}$$

where the sizes of the A sets match the teacher and student rollout budgets used during CGPO. The independent B sets provide two calibration targets. The first is held-out teacher quality,

$$q_T^{B}(x) = \frac{1}{|B_T|}\sum_{k\in B_T} r_{T,k}(x), \tag{5}$$

and the second is the tie-adjusted probability of superiority (Vargha and Delaney, 2000), estimated from all teacher–student pairs,

$$p_{\text{win}}^{B}(x) = \frac{1}{|B_T||B_S|}\sum_{k\in B_T}\sum_{i\in B_S}\left[\mathbb{I}(r_{T,k} > r_{S,i}) + \frac{1}{2}\mathbb{I}(r_{T,k} = r_{S,i})\right], \tag{6}$$

where a tie contributes one half. Thus, $p_{\text{win}}^{B} = 0.5$ indicates balanced empirical teacher wins and losses, whereas values above 0.5 indicate an empirical tendency for the teacher to outperform the student.

We use the affine-sigmoid form commonly employed in held-out post-hoc calibration (Guo et al., 2017). The maps take verifier-based competence estimates as inputs and predict independently

estimated teacher quality and superiority. We fit two monotone one-dimensional maps,

$$\widehat{q}_T^A(x) = \sigma\big(a_T \widehat{c}_T^A(x) + b_T\big)\,, \qquad \widehat{p}_{\text{win}}^A(x) = \sigma\Big(a_\Delta \widehat{\Delta}^A(x) + b_\Delta\Big)\,, \tag{7}$$

by minimizing task-balanced soft-label binary cross-entropy,

$$\mathcal{L}_{\text{cal}} = \sum_{x \in \mathcal{D}_{\text{cal}}} w_x \left[\ell_{\text{BCE}}\big(q_T^B(x), \widehat{q}_T^A(x)\big) + \ell_{\text{BCE}}\big(p_{\text{win}}^B(x), \widehat{p}_{\text{win}}^A(x)\big)\right], \tag{8}$$

where $\ell_{\text{BCE}}(u, v) = -u \log v - (1-u)\log(1-v)$ and $w_x \propto 1/N_{\tau(x)}$ equalizes the aggregate contribution of task families, with $N_{\tau(x)}$ denoting the number of calibration samples from the task family of x. The slopes are constrained to be positive, and repeated independent partitions reduce sensitivity to a particular split. This calibration requires only offline inference and does not update either the teacher or the student. Once estimated, $a_T, b_T, a_\Delta, b_\Delta$ are fixed throughout CGPO.

Calibrated competence routing. During CGPO, the frozen maps are evaluated using the cached teacher competence and current online student competence:

$$\widehat{q}_T(x) = \sigma(a_T c_T(x) + b_T)\,, \qquad \widehat{p}_{\text{win}}(x) = \sigma(a_\Delta \Delta(x) + b_\Delta)\,. \tag{9}$$

Here, $\widehat{q}_T(x)$ estimates the teacher's independent outcome quality, whereas $\widehat{p}_{\text{win}}(x)$ estimates how reproducibly the teacher outperforms the current student. We define the sample-wise routing coefficient as

$$\boxed{\alpha(x) = \text{sg}\big[\widehat{q}_T(x)\,(2\widehat{p}_{\text{win}}(x) - 1)_+\big]}, \tag{10}$$

where $(z)_+ = \max(z, 0)$ and sg stops gradients through the routing decision. The excess-win term is zero whenever the teacher is no more likely than the student to succeed and increases from zero to one as the calibrated win probability rises from 0.5 to 1. Multiplication by $\widehat{q}_T(x)$ further suppresses teachers that beat the student only because both policies perform poorly. Consequently, $\alpha(x) \in [0, 1]$ acts as a calibrated soft conjunction of absolute teacher quality and reproducible marginal advantage.

Routing operates at the sample level; local teacher errors can remain in samples with high aggregate rewards. The contribution and observed behavior of competence routing are examined in Sec. 5.8 and Fig. 5.

On-policy token-level supervision. For a student-generated response y_S, OPD evaluates the teacher along the trajectory visited by the current policy. At decoding step t, the corresponding student prefix is $h_t = (x, q, y_{S,<t})$, and the token-level distillation objective is defined as

$$\mathcal{L}_{\text{OPD}}(x, y_S) = \frac{1}{|y_S|} \sum_{t=1}^{|y_S|} D_{\text{KL}}(p_\theta(\cdot \mid h_t) \,\|\, p_T(\cdot \mid h_t))\,, \tag{11}$$

where $D_{\text{KL}}(p_\theta \| p_T)$ is the reverse KL divergence from the student token distribution to the teacher token distribution, a direction studied for generative-model distillation in MiniLLM and GKD (Gu et al., 2024; Agarwal et al., 2024). The teacher remains frozen during OPD, and gradients are taken only with respect to θ. Since the teacher is queried on student-generated prefixes, the resulting supervision is aligned with the states visited by the evolving policy rather than with an independently generated teacher trajectory.

Joint optimization. The final CGPO objective is

$$\mathcal{L}_{\mathrm{CGPO}}(x) = \mathcal{L}_{\mathrm{GRPO}}(x) + \lambda\, \alpha(x)\, \mathcal{L}_{\mathrm{OPD}}(x), \tag{12}$$

where λ controls the overall strength of teacher supervision. We do not include a reference-policy KL penalty in $\mathcal{L}_{\mathrm{GRPO}}$; the reverse KL in $\mathcal{L}_{\mathrm{OPD}}$ is used for teacher distillation. GRPO remains active throughout the CGPO stage and provides outcome-level optimization, while OPD contributes dense token-level supervision according to the competence-aware routing coefficient.

The resulting optimization exhibits an adaptive transition of supervision across training. For samples where the teacher initially has high calibrated quality and a reproducible advantage, CGPO assigns greater weight to OPD and exploits the teacher's fine-grained token-level knowledge. As the student improves, $c_S(x)$ increases, $\Delta(x)$ decreases, and the calibrated win probability approaches or falls below 0.5, automatically suppressing distillation. Conversely, samples that remain substantially beyond the student's competence continue to receive teacher guidance in proportion to the teacher's independently calibrated quality. CGPO therefore follows the evolving teacher–student competence boundary throughout optimization, replacing a static distillation schedule with sample-adaptive supervision.

5 Evaluation

We conduct a comprehensive empirical evaluation of PolyOCR across diverse text-centric visual intelligence scenarios. The evaluation examines the model's ability to generalize beyond isolated OCR tasks, covering text perception, spatially grounded understanding, structured document analysis, information extraction, and OCR-oriented reasoning. We further investigate the effectiveness of the proposed Competence-Guided Policy Optimization (CGPO) framework.

5.1 Evaluation Protocol

We establish a unified evaluation protocol to assess PolyOCR across diverse OCR and document intelligence scenarios. All compared models are evaluated under consistent benchmark versions, inference configurations, and scoring procedures to ensure reliable and reproducible comparisons.

Models. We evaluate two model variants, PolyOCR-2B and PolyOCR-9B. Both variants adopt the Qwen3.5 vision-language backbone and are trained under the unified OCR training framework described in Sections 3 and 4. Unless otherwise specified, all evaluations are conducted under the same inference configuration, including prompt templates, image preprocessing strategies, visual token budgets, and decoding settings. Additionally, we include PolyOCR-2.7B in the MDPBench evaluation to assess multilingual document parsing capabilities. This variant features a specialized architecture comprising a 1.2B-parameter ViT vision encoder and a 1.5B-parameter Qwen2.5 text decoder, optimized for diverse writing systems and acquisition conditions.

OCRBench v2.1 protocol. We refine the original OCRBench v2 evaluation protocol to improve the reliability, consistency, and discriminative capability of benchmark evaluation. Due to the limited quality of some original reference annotations and the insufficient alignment between certain evaluation metrics and the actual task objectives, we revise two key aspects of the protocol: reference annotation quality and task-specific evaluation criteria. Specifically, reference annotations are systematically audited and corrected to reduce annotation noise, while task-specific evaluation procedures are refined to better align scoring mechanisms with the semantic objectives of individual tasks.

The resulting OCRBench v2.1 protocol preserves the original task coverage while providing more reliable annotations and more task-aligned evaluation criteria. Detailed modifications to annotations, metrics, and evaluation procedures are presented in Appendix E. All OCRBench v2.1 results reported in this work are obtained using the refined evaluation protocol.

Benchmarks and baselines. The evaluation suite consists of five representative benchmarks: OCRBench v2.1 (our revision of OCRBench v2 (Fu et al., 2025)), CC-OCR (Yang et al., 2025), OmniDocBench v1.6 (Ouyang et al., 2025), MDPBench (Li et al., 2026b), and an in-house KIE benchmark. OCRBench v2.1 and CC-OCR provide comprehensive evaluations across diverse OCR scenarios, covering broad text-centric visual capabilities. OmniDocBench v1.6 evaluates document-centric parsing ability, including layout analysis, table understanding, formula recognition, and document-level content reconstruction. MDPBench further extends the evaluation to multilingual document parsing, covering diverse language/script groups and both digital and photographed documents. The in-house KIE benchmark evaluates the model's ability to extract structured information from document images, focusing on field-level information extraction under practical application scenarios.

We compare PolyOCR against a diverse set of representative vision-language models, including both general-purpose and OCR-specialized models with different scales and architectural designs. For public benchmarks, we use their official test sets and retain the corresponding evaluation protocols for CC-OCR, OmniDocBench v1.6, and MDPBench. For OCRBench v2.1, all comparisons are conducted using the refined annotations and scoring procedures introduced above; therefore, scores obtained under the original OCRBench v2 protocol are not directly comparable. For the in-house KIE benchmark, we evaluate models using the same test set and field-level F1 metric to ensure consistent comparison across different systems.

5.2 Overall Comparison

Table 3 summarizes the cross-benchmark performance of PolyOCR against representative general-purpose vision-language models and OCR-specialized systems. The evaluated benchmarks collectively assess a broad spectrum of text-centric visual capabilities, including fundamental text perception, document-level structural understanding, and OCR-oriented semantic reasoning. PolyOCR-9B demonstrates competitive and consistent performance across all evaluated benchmarks. It achieves scores of **80.42** and **78.37** on the English and Chinese settings of OCRBench v2.1, respectively, and obtains a score of **82.43** on CC-OCR. Furthermore, PolyOCR-9B reaches **91.57** on OmniDocBench v1.6, demonstrating its capability to extend from fine-grained text recognition toward complex document structure understanding and parsing.

The cross-benchmark evaluation demonstrates that the proposed model maintains robust performance across heterogeneous OCR scenarios and task formulations. Such capability consistency reflects the effectiveness of the unified training paradigm in developing a general-purpose OCR foundation model with broad text-centric visual understanding abilities. Its competitive performance across diverse task settings, including text transcription, spatial localization, structured document parsing, and semantic reasoning, highlights the model's ability to address varied OCR requirements while maintaining balanced capabilities across different levels of text-centric visual intelligence. The following sections further analyze the model performance on individual benchmarks and provide detailed comparisons across different OCR capabilities.

Table 3 Cross-benchmark comparison of PolyOCR with representative vision-language models. OCRBench v2.1 reports English/Chinese scores. Bold and underline indicate the best and second-best results, respectively. "–" indicates models not evaluated on the corresponding benchmark in this report.

Model	OCRBench v2.1 (en/zh) ↑	CC-OCR ↑	OmniDocBench v1.6 ↑
General-purpose Vision-Language Models			
Qwen3.5-2B	54.03/63.75	65.66	79.95
Qwen3.5-9B	66.64/71.88	71.44	89.49
Qwen3-VL-2B-Instruct	44.27/48.84	51.02	80.05
Qwen3-VL-8B-Instruct	54.36/70.56	70.29	78.94
GLM-4.6V-Flash	67.89/75.34	72.50	71.42
InternVL3.5-8B	51.09/62.88	65.88	75.89
Qwen3.6-35B-A3B	70.34/76.25	73.19	89.01
Qwen3.5-122B-A10B	70.34/<u>78.77</u>	73.79	89.04
GPT-5.6-Sol	72.92/67.80	74.54	88.00
Qwen3.8-Max	<u>77.75</u>/**79.32**	73.64	<u>92.31</u>
OCR-specialized Vision-Language Models			
HunyuanOCR	32.26/54.44	54.57	87.92
HunyuanOCR-1.5	32.63/49.41	51.10	92.17
Qianfan-OCR	57.09/70.87	<u>82.06</u>	88.37
OvisOCRv2	25.67/50.28	36.26	**96.58**
PolyOCR (Ours)			
PolyOCR-2B	72.33/71.32	80.53	88.12
PolyOCR-9B	**80.42**/78.37	**82.43**	91.57

5.3 OCRBench v2.1 Evaluation

OCRBench v2.1 provides a comprehensive evaluation of text-centric visual intelligence across diverse OCR capabilities, including text recognition, localization, spotting, document parsing, formula understanding, translation, and OCR-oriented reasoning. Table 4 presents a detailed comparison across these capability categories, providing a fine-grained assessment of model performance beyond overall accuracy.

PolyOCR-9B achieves competitive performance on OCRBench v2.1, obtaining overall scores of **80.42** and **78.37** on the English and Chinese settings, respectively. The model demonstrates strong results across text perception, document understanding, and OCR-oriented reasoning. These results indicate that the proposed unified training framework effectively enables the model to handle heterogeneous OCR tasks within a single generative modeling paradigm.

At the category level, PolyOCR-9B ranks first among the compared models in five of the 13 language-specific categories: English text recognition (82.99), text detection (70.84), element parsing (89.70), and visual text understanding (87.58), together with Chinese element parsing (86.86). It ranks second in English text spotting (78.50) and English relationship extraction (95.51). These results show strengths in text perception and structured parsing, while mathematical calculation, Chinese text recognition, and Chinese visual text understanding remain below the strongest evaluated baselines.

Overall, the OCRBench v2.1 results demonstrate that PolyOCR provides a balanced and comprehensive OCR capability across heterogeneous task formulations, establishing a strong foundation for unified text-centric visual intelligence.

Table 4 Fine-grained OCRBench v2.1 comparison of PolyOCR with representative vision-language models. TR, TD, TS, RE, EP, MC, VTU, and KR denote Text Recognition, Text Detection, Text Spotting, Relationship Extraction, Element Parsing, Mathematical Calculation, Visual Text Understanding, and Knowledge Reasoning, respectively. The **All** column reports overall English/Chinese scores (en/zh). Bold and underline indicate the best and second-best results.

	English								Chinese					
Model	TR	TD	TS	RE	EP	MC	VTU	KR	TR	RE	EP	VTU	KR	All (en/zh)
General-purpose VLM														
Qwen3.5-2B	67.44	31.48	8.50	80.13	58.73	52.65	71.73	61.58	73.48	71.67	66.15	60.50	46.95	54.03/63.75
Qwen3.5-9B	72.16	54.42	49.94	89.44	66.12	60.45	76.23	64.37	76.42	78.62	67.35	65.50	71.52	66.64/71.88
Qwen3-VL-2B-Instruct	51.83	24.09	0.64	73.49	49.58	29.89	70.44	54.19	46.55	67.90	50.81	50.00	28.93	44.27/48.84
Qwen3-VL-8B-Instruct	67.71	26.94	4.20	88.54	64.76	45.11	74.85	62.79	77.75	83.91	72.26	68.00	50.87	54.36/70.56
GLM-4.6V-Flash	78.04	54.08	38.02	83.21	68.13	71.10	82.14	68.44	76.78	84.53	77.60	62.00	75.79	67.89/75.34
InternVL3.5-8B	60.96	27.45	5.89	83.79	54.12	46.25	70.95	59.33	63.00	70.38	69.17	53.00	58.84	51.09/62.88
Qwen3.6-35B-A3B	75.15	58.32	67.13	92.36	64.23	61.83	77.13	66.58	75.60	85.95	72.35	66.00	81.34	70.34/76.25
Qwen3.5-122B-A10B	78.63	58.53	62.24	95.22	68.96	59.17	76.49	63.50	80.39	87.07	79.65	66.50	80.25	70.34/78.77
GPT-5.6-Sol	74.21	61.48	67.45	88.55	65.87	76.73	76.59	72.46	70.10	71.90	74.06	51.00	71.95	72.92/67.80
Qwen3.8-Max	81.52	70.04	**78.62**	**96.58**	69.36	75.47	77.23	**73.16**	80.73	**89.86**	73.98	68.00	84.04	77.75/**79.32**
Doubao-Seed-2.0-Lite	71.35	66.91	62.17	90.62	65.03	**82.66**	83.43	72.12	59.89	83.34	71.43	62.50	**93.40**	74.28/74.11
Gemini-3.5-Flash-Lite	72.71	24.94	0.50	92.30	65.23	49.22	76.22	72.20	37.74	79.66	54.61	66.50	51.51	56.67/58.01
Kimi-K3	77.20	57.25	45.77	91.65	65.02	58.93	75.88	70.56	62.36	80.24	67.07	62.50	81.59	67.78/70.75
OCR-specialized VLM														
HunyuanOCR	72.26	5.50	0.00	19.84	44.00	12.25	63.11	41.13	75.86	62.59	60.18	48.81	24.76	32.26/54.44
HunyuanOCR-1.5	71.77	14.99	0.33	1.11	48.16	14.55	65.39	44.70	75.73	21.98	61.26	55.29	32.79	32.63/49.41
Qianfan-OCR	73.56	40.04	1.86	90.82	77.42	31.84	81.53	59.70	**80.78**	85.24	86.09	62.50	39.74	57.09/70.87
OvisOCRv2	57.19	0.00	0.00	0.85	43.31	17.90	53.93	32.14	73.00	29.72	65.89	**69.00**	13.81	25.67/50.28
PolyOCR (Ours)														
PolyOCR-2B	77.90	60.90	60.11	94.06	89.30	53.09	81.47	61.79	77.26	79.39	86.34	52.50	61.09	72.33/71.32
PolyOCR-9B	**82.99**	**70.84**	78.50	95.51	**89.70**	68.15	**87.58**	70.11	77.13	85.38	**86.86**	60.50	82.00	**80.42**/78.37

5.4 CC-OCR Evaluation

CC-OCR provides an evaluation of practical OCR capabilities under diverse visual conditions, covering real-world text reading, document-level understanding, structured information extraction, and multilingual OCR. Our CC-OCR evaluation covers Multi-Scene OCR, Document Parsing, and Key Information Extraction (KIE); the multilingual track is excluded.

Table 5 presents the comparison across the three evaluated tracks. PolyOCR-9B achieves an overall score of **82.43**, demonstrating competitive performance across practical OCR scenarios. In particular, the model obtains the best results on Document Parsing and Key Information Extraction, reaching scores of **71.71** and **94.02**, respectively. These results indicate that the proposed unified training framework effectively supports both fine-grained text understanding and document-level structured prediction.

The strong performance on KIE further highlights the advantage of jointly modeling text perception and semantic information extraction within a unified OCR framework. PolyOCR learns to associate visual text regions with semantic structures and output formats, supporting structured information extraction from complex documents. Together with the Document Parsing results, the CC-OCR evaluation demonstrates the capability of PolyOCR to extend beyond text recognition toward structured text-centric visual intelligence.

5.5 OmniDocBench v1.6 Evaluation

OmniDocBench v1.6 targets document parsing within complex page layouts, focusing on the evaluation of capabilities such as text recognition, formula recognition, and table parsing. Table 6

Table 5 CC-OCR results of PolyOCR and representative general-purpose and OCR-specialized vision-language models. The overall score is the average of Document Parsing, Key Information Extraction, and Multi-Scene OCR. Bold and underline indicate the best and second-best results, respectively.

Model	Doc Parsing ↑	KIE ↑	Multi-Scene OCR ↑	Overall ↑
General-purpose Vision-Language Models				
Qwen3.5-2B	32.48	83.01	81.49	65.66
Qwen3.5-9B	37.95	91.40	84.98	71.44
Qwen3-VL-2B-Instruct	16.38	81.48	55.21	51.02
Qwen3-VL-8B-Instruct	38.89	91.81	80.16	70.29
GLM-4.6V-Flash	49.43	83.12	84.94	72.50
InternVL3.5-8B	35.92	86.96	74.75	65.88
Qwen3.6-35B-A3B	41.46	92.31	85.80	73.19
Qwen3.5-122B-A10B	41.71	<u>93.72</u>	<u>85.94</u>	73.79
GPT-5.6-Sol	51.42	89.33	82.87	74.54
Qwen3.8-Max	42.68	91.12	**87.11**	73.64
OCR-specialized Vision-Language Models				
HunyuanOCR	40.95	45.34	77.41	54.57
HunyuanOCR-1.5	68.63	0.55	84.13	51.10
Qianfan-OCR	71.18	92.12	82.88	<u>82.06</u>
OvisOCRv2	39.16	0.00	69.64	36.26
PolyOCR (Ours)				
PolyOCR-2B	<u>71.50</u>	90.90	79.18	80.53
PolyOCR-9B	**71.71**	**94.02**	81.55	**82.43**

summarizes PolyOCR's performance across various document components and reading orders. When processing full pages directly, PolyOCR achieved a composite score of **91.57**, demonstrating strong competitiveness compared to general-purpose vision-language models. This indicates that the recognition capabilities acquired through unified OCR training can be applied to page-level document parsing, enabling the model to extract diverse types of document content from complex layouts.

To further examine PolyOCR's parsing capabilities regarding various document components, we introduced a layout-guided pipeline setup. This configuration utilizes PP-DocLayoutV3 to detect layout elements and predict reading order; the same PolyOCR model then parses the cropped regions containing texts, formulas, and tables, and the results are finally assembled into a Markdown document based on the predicted reading order. By delegating page layout analysis to an external module, this setup allows PolyOCR to focus more effectively on content recognition and structural recovery within the specific regions.

Under this configuration, the composite score rose from **91.57** to **95.52**, with improvements observed across all reported metrics. Notably, the formula recognition CDM score increased from 91.71 to 96.30, and the table parsing TEDS score improved from 88.67 to 93.56. These results further validate PolyOCR's effectiveness as a unified document component parser capable of handling texts, formulas, and tables with a single model. When combined with a layout analysis module, PolyOCR achieves page-level parsing performance comparable to leading specialized systems.

5.6 In-house KIE Benchmark Evaluation

We evaluate PolyOCR on an in-house KIE benchmark covering four image types: Xuexin Network certificates, identity cards, admission notices, and international passports. We report field-level F1 scores for each image type as well as the overall test set performance.

Table 6 Fine-grained OmniDocBench v1.6 results of PolyOCR and representative general-purpose and OCR-specialized vision-language models. Bold and underline indicate the best and second-best results, respectively.

	Text^{Edit} ↓	Formula^{CDM} ↑	Table^{TEDS} ↑	Table$^{\text{TEDS-S}}$ ↑	Read Order^{Edit} ↓	Overall ↑
General-purpose VLM						
Qwen3.5-2B	0.149	86.92	67.79	72.60	0.231	79.95
Qwen3.5-9B	0.080	92.44	84.01	87.42	0.178	89.49
Qwen3-VL-8B-Instruct	0.110	79.68	68.20	73.69	0.222	78.94
GLM-4.6V-Flash	0.238	77.96	60.14	64.07	0.280	71.42
InternVL3.5-8B	0.175	79.18	66.02	72.20	0.241	75.89
Qwen3.6-35B-A3B	0.077	91.35	83.41	86.18	0.176	89.01
Qwen3.5-122B-A10B	0.063	92.23	81.18	84.06	0.161	89.04
GPT-5.6-Sol	0.109	89.68	85.18	88.87	0.165	88.00
Qwen3.8-Max	0.054	94.49	87.81	90.65	0.138	92.31
OCR-specialized VLM						
HunyuanOCR	0.073	87.15	83.92	86.62	0.152	87.92
HunyuanOCR-1.5	0.049	89.95	91.50	93.05	0.131	92.17
Qianfan-OCR	0.105	92.68	82.96	85.70	0.168	88.37
MinerU2.5-Pro	0.036	97.45	93.42	95.92	<u>0.120</u>	95.75
PaddleOCR-VL-1.6	0.033	<u>97.49</u>	<u>94.76</u>	97.11	0.127	96.33
OvisOCRv2	**0.025**	**97.53**	<u>94.76</u>	<u>97.16</u>	**0.111**	<u>96.58</u>
NaviDC-OCR	<u>0.027</u>	96.36	**97.05**	**98.52**	0.122	**96.87**
PolyOCR (Ours)						
PolyOCR-9B	0.057	91.71	88.67	91.80	0.139	91.57
PolyOCR-9B(Pipeline)	0.033	96.30	93.56	95.38	0.123	95.52

As shown in Table 7, PolyOCR-9B achieves the highest overall F1 score of **93.60%** among all evaluated models. It achieves the best performance on admission notices, reaching an F1 score of **88.35%**. On Xuexin Network certificates, PolyOCR-9B obtains an F1 score of 95.40%, slightly below PolyOCR-2B (95.53%), while both PolyOCR variants outperform all other evaluated models. For identity cards, PolyOCR-9B achieves an F1 score of 99.20%, which is competitive with the best-performing model, Qwen3.6-35B-A3B (99.53%). On international passports, PolyOCR-9B reaches an F1 score of 93.80%, ranking second among the evaluated models and trailing GPT-5.6-Sol (98.55%) by 4.75 percentage points.

Overall, PolyOCR demonstrates strong and balanced performance across the four image types. The PolyOCR variants achieve leading results on the benchmark, with PolyOCR-9B showing particularly strong capabilities on admission notices and maintaining competitive performance on other complex KIE scenarios. Moreover, the overall F1 score of 93.60% achieved by PolyOCR-9B surpasses those of the evaluated larger-scale models, including GPT-5.6-Sol (90.98%), Qwen3.6-35B-A3B (89.17%), and Qwen3.5-122B-A10B (89.03%). These results demonstrate the effectiveness of PolyOCR in handling diverse KIE scenarios and highlight its potential for practical document information extraction applications.

5.7 MDPBench Evaluation

MDPBench (Li et al., 2026b) evaluates multilingual document parsing across digital and photographed documents, covering 17 language/script groups with diverse writing systems and acquisition conditions. The benchmark uses $1 - \text{NED}$ for text recognition and reading order, where NED denotes normalized edit distance, and CDM and TEDS for formula and table recognition, respectively. Scores are aggregated at the page level before averaging across pages. Table 8 reports overall

Table 7 Field-level F1 scores (%) on the in-house KIE benchmark. The benchmark contains four image types: Xuexin Network certificates, identity cards, admission notices, and international passports. The highest value in each column is shown in bold, and the second-highest value is underlined.

Model	Xuexin	Identity Card	Admission Notice	Passport	Overall
GPT-5.6-Sol	93.03	96.31	79.58	**98.55**	90.98
Qwen3.6-35B-A3B	89.19	**99.53**	82.49	89.54	89.17
Qwen3.5-122B-A10B	92.33	<u>99.43</u>	79.27	88.74	89.03
GLM-4.6V-Flash	90.83	98.94	74.35	56.18	79.71
Qianfan-OCR	90.79	99.38	69.98	59.78	78.81
InternVL3.5-8B	87.47	96.58	61.57	57.37	74.66
HunyuanOCR	62.75	99.38	60.13	31.34	61.92
PolyOCR-2B	**95.53**	99.17	<u>88.17</u>	93.41	<u>93.49</u>
PolyOCR-9B	<u>95.40</u>	99.20	**88.35**	<u>93.80</u>	**93.60**

performance, acquisition-type breakdowns, and both group-level and individual language/script scores. All scores are expressed as percentages, with higher values indicating better performance.

To assess multilingual document parsing within the PolyOCR family, we additionally evaluate PolyOCR-2.7B, a specialized configuration combining a 1.2B-parameter ViT vision encoder with a 1.5B-parameter Qwen2.5 text decoder. The model is trained on an in-house document corpus curated through cross-model consistency checks, multilingual quality filtering, and validation of fields and reading order.

As shown in Table 8, PolyOCR-2.7B leads four of the five summary measures and ties for the highest score on photographed documents among the listed methods. It achieves an overall score of 87.0%, exceeding GPT-5.6-Sol and Gemini-3-Pro by 0.4 and 0.6 percentage points, respectively. On digital documents, it reaches 91.0%, surpassing the strongest listed baseline, Kimi-K3, by 0.2 percentage points. On photographed documents, it obtains 85.6%, matching GPT-5.6-Sol. Together, these results show competitive performance across both acquisition types.

Compared with the listed OCR-specialized models, PolyOCR-2.7B achieves higher scores across all five summary measures. Relative to MonkeyOCRv2-B-Parsing, the OCR-specialized baseline with the highest overall score, it improves overall performance by 3.7 percentage points and photographed-document performance by 3.9 percentage points. The corresponding gains on the Latin and non-Latin averages are 4.6 and 2.8 percentage points, respectively, indicating that its advantage extends across both language/script groups.

The finer-grained results show Latin and non-Latin averages of 88.8% and 84.9%, respectively, each exceeding the strongest corresponding reference result by 0.4 percentage points. At the individual language/script level, PolyOCR-2.7B achieves the highest scores in six groups—Spanish, French, Dutch, Vietnamese, Japanese, and Traditional Chinese—and ranks among the top two in 12 of the 17 groups. Overall, these results demonstrate strong multilingual document parsing performance for the 2.7B configuration across diverse writing systems and acquisition conditions.

5.8 Ablation Study of CGPO

The preceding results establish the overall capability of PolyOCR. We next compare post-training variants to assess Competence-Guided Policy Optimization. All variants are initialized from the same SFT checkpoint.

The ablation considers five post-training configurations: GRPO alone, OPD alone, GRPO combined

Table 8 Fine-grained MDPBench scores (%) of PolyOCR, general-purpose VLMs, OCR-specialized VLMs, and document parsing pipelines. All denotes the overall score; Digit. and Photo. report scores on digital and photographed documents, respectively. Latin and Non-Latin include group averages (Avg.) and individual language/script scores. Higher scores indicate better performance. Bold and underline indicate the highest and second-highest distinct scores, respectively, in each column across all models and tools shown. Tied scores receive the same formatting.

Model	Overall			Latin										Non-Latin								
	All	Digit.	Photo.	Avg.	DE	EN	ES	FR	ID	IT	NL	PT	VI	Avg.	AR	HI	JP	KO	RU	TH	ZH	ZH-T
General-purpose Vision-Language Models																						
GPT-5.6-Sol	86.6	90.2	**85.6**	86.7	88.5	89.2	82.1	81.4	88.7	89.1	85.6	88.8	86.9	84.5	88.2	88.6	76.5	84.8	**86.2**	**81.2**	86.4	84.4
Gemini-3-Pro	86.4	90.4	85.1	88.4	**91.2**	90.6	83.4	82.7	**91.5**	91.6	87.7	**91.4**	85.9	84.1	89.4	**90.4**	74.8	85.5	84.9	80.6	85.1	82.1
Kimi-K3	83.6	90.8	81.2	86.2	89.1	87.2	80.2	80.0	86.9	**92.7**	86.0	88.9	84.8	80.7	77.4	77.5	74.9	**89.9**	82.4	72.1	**89.5**	81.9
Qwen3-VL-Instruct-8b	68.3	78.4	65.0	73.6	73.7	71.4	69.3	66.2	68.5	79.1	78.3	82.2	73.4	62.5	63.1	58.4	59.9	61.9	57.9	62.0	62.6	73.8
Qwen3.5-Instruct-9B	65.7	74.8	62.7	72.5	72.8	72.0	72.0	64.4	66.2	77.6	74.5	79.1	74.0	58.2	53.4	56.2	55.7	60.3	54.7	56.7	60.8	67.5
InternVL-3.5-8B	42.7	59.7	37.0	53.4	39.8	64.2	47.5	42.7	53.8	60.6	52.2	63.2	57.0	30.6	8.2	9.0	45.6	30.3	26.1	10.8	55.3	59.3
OCR-specialized Vision-Language Models																						
MonkeyOCRv2-B-Parsing	83.3	88.1	81.7	84.2	87.7	84.5	75.2	78.4	86.5	88.6	86.1	87.9	83.2	82.1	**90.7**	87.2	71.9	87.6	80.1	80.8	83.6	75.3
dots.mocr	80.5	90.5	77.2	81.7	82.6	87.4	71.3	70.1	84.5	89.3	83.2	86.8	79.9	79.2	83.3	83.6	75.0	78.7	71.2	77.9	84.6	79.6
PaddleOCR-VL-1.6	78.9	87.1	76.3	81.4	85.1	83.5	75.1	78.6	82.7	85.5	80.5	80.9	80.9	76.1	76.2	70.7	70.3	86.3	75.1	67.3	85.9	76.7
HunyuanOCR-1.5	76.8	86.2	73.6	79.7	79.6	80.4	74.2	70.0	81.5	84.5	78.4	86.4	82.4	73.5	71.8	71.6	65.5	75.7	67.4	77.7	80.8	77.2
MinerU-2.5-Pro-1.2B	71.0	86.2	66.1	74.6	78.3	79.5	63.4	67.4	78.0	79.7	72.1	78.6	74.2	67.0	56.6	72.2	59.1	77.6	62.6	61.8	76.5	69.7
PaddleOCR-VL	69.6	87.6	63.6	72.1	78.2	79.3	62.9	66.0	77.4	78.4	67.9	72.0	66.6	66.7	65.8	68.4	59.9	77.8	56.9	57.8	78.2	68.5
OvisOCR2	69.3	83.7	64.5	80.2	80.4	**90.9**	75.6	67.4	79.9	87.1	82.8	85.6	71.9	56.9	45.8	43.8	69.9	64.6	57.0	15.1	81.2	77.7
HunyuanOCR	68.3	80.2	64.3	72.4	75.0	73.1	63.0	66.1	69.9	80.3	61.4	81.9	80.6	63.7	68.3	73.1	55.6	68.9	52.2	60.7	66.8	64.2
GLM-OCR	67.3	77.9	63.7	78.7	82.7	84.5	75.8	76.2	79.7	82.8	80.2	77.4	69.2	54.3	21.7	39.6	65.5	61.2	64.2	27.4	78.5	76.7
Pipeline Tools																						
PP-StructureV3	45.4	56.2	41.7	59.8	60.4	68.7	54.4	49.8	69.6	68.9	55.5	58.4	52.7	28.9	1.0	7.7	56.2	15.4	7.5	11.9	72.2	59.1
MinerU-2.5-pipeline	33.5	57.6	25.4	46.5	54.3	58.3	38.4	43.6	51.9	56.5	43.9	44.0	27.6	18.7	1.2	5.3	24.5	6.8	4.2	6.4	53.9	47.2
PolyOCR (Ours)																						
PolyOCR-2.7B	**87.0**	**91.0**	**85.6**	**88.8**	90.5	90.2	**85.0**	**84.8**	90.0	90.8	**89.2**	90.5	**88.6**	**84.9**	88.3	89.0	**77.2**	86.8	85.4	81.0	85.8	**85.9**

with fixed-weight OPD, GRPO combined with cosine-decayed OPD, and the proposed CGPO. These comparisons examine outcome-based policy optimization, on-policy token-level supervision, and sample-wise competence routing alongside a step-dependent distillation schedule.

Step-dependent baseline. To test whether a predefined decline in distillation strength accounts for the gains from CGPO, the cosine baseline replaces the sample-dependent coefficient $\alpha(x)$ with

$$\alpha_t = \alpha_{\min} + \frac{\alpha_{\max} - \alpha_{\min}}{2}\left(1 + \cos\frac{\pi t}{T}\right), \qquad 0 \le t \le T, \tag{13}$$

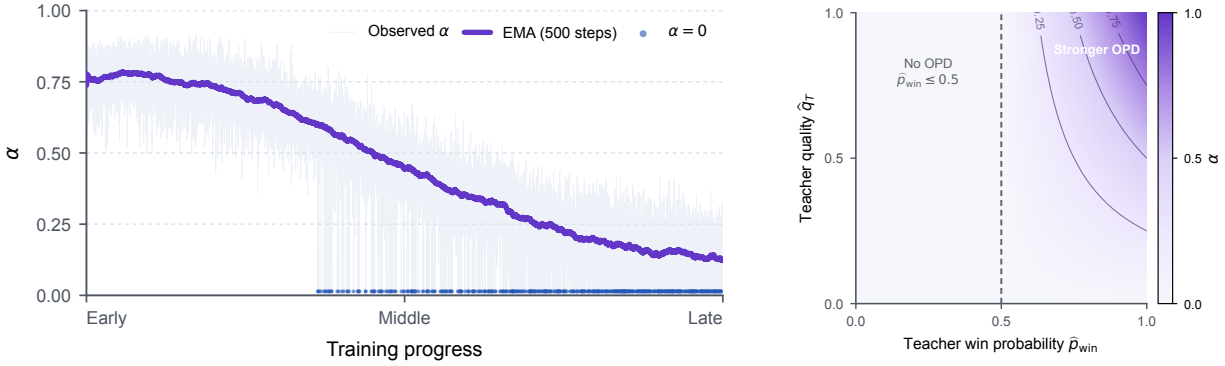
where t is the current optimization step, T is the total number of optimization steps, and $\alpha_{\max}$ and $\alpha_{\min}$ are the initial and final OPD weights, respectively. The joint objective remains $\mathcal{L}_{\text{GRPO}} + \lambda\alpha_t\mathcal{L}_{\text{OPD}}$. At a given step, all samples share the same weight, independent of teacher competence or the teacher–student competence gap.

Component-wise comparison. Table 9 summarizes the results. GRPO provides direct optimization toward task-level outcome rewards and establishes the contribution of policy optimization beyond SFT. OPD introduces additional token-level supervision along student-generated trajectories. Combining GRPO and OPD with a fixed coefficient further improves performance by jointly exploiting outcome-level and token-level signals.

CGPO achieves the highest scores on all three benchmarks among the evaluated post-training strategies. On CC-OCR, it scores 82.43, improving on the SFT checkpoint by 2.97 points and on fixed-weight GRPO–OPD by 0.25 points. Its routing rule assigns stronger distillation weights when the teacher is estimated to provide reliable, non-redundant information.

Table 9 Ablation study of Competence-Guided Policy Optimization. All variants are initialized from the same SFT checkpoint. Cosine decay uses the step-dependent OPD weight in Eq. 13, without competence routing.

Setting	GRPO	OPD	Competence Routing	OCRBench v2.1 (en/zh)	CC-OCR	OmniDocBench
SFT checkpoint				72.08/74.71	79.46	89.72
+ GRPO	✓			74.63/76.24	81.72	90.54
+ OPD		✓		73.84/75.77	80.91	90.69
+ GRPO + OPD	✓	✓		75.42/77.08	82.18	91.16
+ GRPO + OPD (cosine)	✓	✓		78.23/76.31	82.32	90.12
+ CGPO	✓	✓	✓	**80.42/78.37**	**82.43**	**91.57**



(a) **(b)**

Figure 5 CGPO routing behavior. **(a)** Observed routing coefficients during CGPO, their exponential moving average, and points with $\alpha = 0$. **(b)** Analytical routing surface induced by $\alpha = \widehat{q}_T(2\widehat{p}_{\text{win}} - 1)_+$. OPD is disabled when $\widehat{p}_{\text{win}} \le 0.5$ and receives increasing weight as teacher quality and the calibrated probability of teacher superiority increase.

Compared with cosine-decayed OPD, CGPO improves OCRBench v2.1 scores by 2.19 and 2.06 points in English and Chinese, respectively, CC-OCR by 0.11 points, and OmniDocBench by 1.45 points. This comparison shows higher benchmark scores for CGPO than for the evaluated step-dependent schedule.

Routing dynamics. Figure 5 shows the observed routing coefficient during CGPO. Despite local fluctuations, its smoothed trajectory declines from early to late training, and zero-weight decisions become more frequent toward the later portion of the trace. The analytical surface in panel (b) illustrates how teacher quality and the calibrated probability of teacher superiority jointly determine the OPD weight.

6 Conclusion

This report presents PolyOCR, a unified OCR foundation model that advances OCR from isolated text recognition toward comprehensive text-centric visual intelligence. By integrating capability-oriented data construction with competence-aware post-training, PolyOCR establishes a unified modeling paradigm for diverse OCR scenarios, encompassing scene text understanding, document parsing, information extraction, structured content recovery, multilingual transformation, and OCR-oriented reasoning.

The proposed data construction framework provides large-scale, diverse, and quality-controlled

supervision across heterogeneous OCR capabilities by integrating real-world resources, controllable synthesis, and model-assisted annotation. Building upon this foundation, Competence-Guided Policy Optimization (CGPO) further enhances model capabilities through adaptive coordination of outcome-based policy optimization and on-policy distillation. By dynamically allocating teacher supervision according to the evolving competence boundary between teacher and student models, CGPO improves the effectiveness of reinforcement learning and knowledge transfer under heterogeneous OCR tasks.

Extensive evaluations on OCRBench v2.1, CC-OCR, OmniDocBench v1.6, MDPBench, and the in-house KIE benchmark demonstrate that PolyOCR achieves strong and balanced performance across text recognition, localization, structured document understanding, multilingual parsing, and real-world information extraction. These results validate the effectiveness of the proposed unified training framework and highlight the feasibility of developing OCR foundation models capable of generalizing across diverse text-containing visual environments while maintaining competitive performance with specialized OCR systems.

Future work will focus on further strengthening fundamental OCR capabilities, such as fine-grained text perception, localization, and recognition, while advancing semantic understanding and reasoning beyond text extraction. Enhancing the generalization and robustness of OCR foundation models in diverse and complex real-world environments remains a key direction toward more reliable text-centric visual intelligence.

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Appendix

A Contributors

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B Implementation and Reproducibility Details

This appendix records the implementation details and supplementary evaluation protocols for PolyOCR.

B.1 Model and Input Configuration

PolyOCR is released at three parameter scales. All share the same task interface, data engine, and training settings for SFT and CGPO, including input settings and optimization hyperparameters. Table 10 records the base models and input configuration.

Table 10 Model and input configuration. All variants share the same training settings.

Configuration	PolyOCR-2B	PolyOCR-9B	PolyOCR-2.7B
Base checkpoint	Qwen3.5-2B	Qwen3.5-9B (ModelScope copy)	ViT (1.2B) + Qwen2.5 (1.5B)
Image-area setting (`MAX_PIXELS`)	1,048,576 pixels	1,048,576 pixels	1,048,576 pixels
Maximum SFT sequence length	20,480 tokens	20,480 tokens	20,480 tokens
SFT precision	BF16	BF16	BF16

The image-area entry records the environment setting in the SFT launch script.

B.2 Supervised Fine-Tuning Configuration

Supervised fine-tuning jointly trains the model across text recognition, localization, spotting, key information extraction, document and table parsing, formula recognition, translation, and OCR-centric reasoning.

Both PolyOCR-2B and PolyOCR-9B use the same SFT configuration: ms-swift (`swift sft`) with DeepSpeed ZeRO-2 on 16 nodes with 16 GPU workers per node. We update the language-model parameters in full-tuning mode while freezing the vision encoder and multimodal aligner. Training uses BF16 and the `flash_attention_3` attention implementation. The model and prompt-template types are both `qwen3_5`; non-thinking prefixes are enabled, and empty thinking spans are excluded

Within each group, names are ordered by first name in alphabetical order.

from the loss through `ignore_empty_think`. The training mixture follows the 60M-instance, three-phase schedule in Section 4.2.

Table 11 Shared supervised fine-tuning configuration for PolyOCR-2B and PolyOCR-9B.

Configuration	Value
Effective sampled instances	Approximately 60M; three phases of 20M
Maximum SFT sequence length	20,480 tokens
Task-mixture weights (TP/DS/OR/GA)	$32.5/30.0/22.5/15.0\%$
GPU workers	256 (16 nodes $\times$ 16 workers)
Per-device training batch size	1
Gradient accumulation steps	1
Global training batch size	256
Number of epochs	1
Peak learning rate and schedule	10^{-5}; cosine decay
Warm-up ratio	5%
Length-based grouping	Enabled
Vision-encoder gradient checkpointing	Disabled
Checkpoint saving interval	2,000 steps
Checkpoint retention	4; model weights only
Logging interval	10 steps
Data-loader workers per process	12
Dataset preprocessing processes	128

The global training batch size is $256 \times 1 \times 1 = 256$. Training is configured for one epoch.

B.3 Competence-Guided Policy Optimization Configuration

All model variants use the same CGPO configuration. Each starts from its corresponding SFT checkpoint and trains with a learning rate of 10^{-6} and Qwen3.5-122B-A10B as the frozen teacher. Teacher competence is estimated from $K = 8$ offline generations per sample and cached, whereas student competence is updated from $G = 8$ rollouts per group. Both estimates use the same quality-verified reference and task-specific verifier. Table 12 summarizes the setup defined in Section 4.3.

Before policy optimization, we fit the teacher-quality and win-probability calibration maps on a fixed split stratified by task and difficulty. Independent A/B partitions of teacher and SFT-student generations separate the router inputs from the calibration targets. For each calibration sample, $|A_T| = |A_S| = 8$ matches the competence-estimation budgets used during CGPO, and the independent target partitions contain $|B_T| = |B_S| = 8$ generations. Calibration uses task-balanced soft-label binary cross-entropy with positive slopes and repeated independent partitions. The fitted coefficients $a_T, b_T, a_\Delta, b_\Delta$ remain fixed throughout policy optimization.

Table 12 Shared CGPO setup and calibrated routing for all model variants.

Configuration	Setting
Student initialization	SFT checkpoint
Teacher model	Qwen3.5-122B-A10B; frozen
Learning rate	10^{-6}
Rollout sampling temperature	0.9
GRPO clipping	$\epsilon = 0.2$; symmetric ratio clipping to $[0.8, 1.2]$
Reference-policy KL penalty	Disabled ($\beta = 0$)
Teacher competence $c_T(x)$	Mean verifier reward over $K = 8$ offline generations per sample; cached
Student competence $c_S(x)$	Mean verifier reward over $G = 8$ current student rollouts per group
Calibration split	Fixed training split stratified by task and difficulty
Calibration A budgets	$\lvert A_T\rvert = 8$, $\lvert A_S\rvert = 8$ per calibration sample
Calibration B budgets	$\lvert B_T\rvert = 8$, $\lvert B_S\rvert = 8$ per calibration sample; independent of A
Calibration targets	Held-out teacher quality q_T^B and pairwise superiority p_{win}^B; ties contribute 0.5
Calibration maps	Positive-slope sigmoid maps for teacher quality and teacher–student competence gap
Calibration fitting	Task-balanced soft-label BCE, $w_x \propto 1/N_{\tau(x)}$; repeated independent partitions
Calibration coefficients	Fitted offline; $a_T, b_T, a_\Delta, b_\Delta$ frozen during CGPO
Routing coefficient	$\alpha(x) = \text{sg}[\widehat{q}_T(x)(2\widehat{p}_{\text{win}}(x) - 1)_+]$
Zero OPD-weight condition	$\widehat{p}_{\text{win}}(x) \le 0.5$
OPD objective	Length-averaged reverse KL, $D_{\text{KL}}(p_\theta \Vert p_T)$, on student-generated prefixes
OPD coefficient λ	1
Joint optimization	$\mathcal{L}_{\text{GRPO}}(x) + \lambda\alpha(x)\mathcal{L}_{\text{OPD}}(x)$

The rollout temperature and clipping coefficient follow the ms-swift GRPO defaults. We disable the reference-policy KL penalty ($\beta = 0$); the reverse KL in OPD is retained for teacher distillation. GRPO remains active throughout CGPO, with $\lambda = 1$ in the joint objective. The stopped-gradient coefficient $\alpha(x)$ controls the contribution of OPD; when the calibrated teacher superiority is at most 0.5, the OPD weight is zero. During policy optimization, gradients update the student while the teacher and calibration maps remain frozen.

C Task Interfaces and Output Normalization

All task families use autoregressive conditional generation while retaining task-appropriate output representations and verifiers. Table 13 summarizes the canonical representations used for training and evaluation. Exact serialization grammars, prompt templates, and normalization code will be released with the model and evaluator.

Table 13 Task interfaces, output normalization, and scalar verifier rewards.

Task family	Canonical output	Normalization / validity guard	Scalar reward
Text recognition	Plain-text transcription	Unicode and whitespace normalization; empty or truncated output is invalid	NED similarity
Grounding	One or more bounding boxes	Coordinate-range validation and one-to-one box assignment	IoU
Text spotting	Box–text pairs	Joint schema validation, box matching, and transcription normalization	IoU–NED geometric mean
KIE	Schema-constrained key–value records	Key canonicalization, type-aware value normalization, duplicate-key rejection	Soft field-level F1
Full-page parsing	Reading-ordered structured text	Element parsing and availability-aware component extraction	NED / CDM / TEDS aggregation
Table extraction	HTML-like table structure	HTML normalization and structural parse validation	TEDS
Formula recognition	LaTeX expression	Syntax normalization followed by rendering validation	CDM
Translation	Translated text	Source-target language validation and deterministic judge schema	LLM-judge score
OCR-centric VQA	Short or open-form answer	Task-specific answer normalization and deterministic judge schema	Exact match or LLM judge
Counting / rule-based reasoning	Answer with optional structured rationale	Output-schema validation and answer recomputation where possible	Rule-based accuracy

C.1 Normalization Order

Verifier preprocessing follows a fixed order: output parsing, schema validation, task-specific canonicalization, reference-prediction matching, and scalar scoring. Parsing failures and out-of-schema outputs receive zero reward. Canonicalization removes representational differences that do not change the task answer, but does not repair missing fields, hallucinated content, invalid coordinates, or incorrect structure. For all rule-based verifiers, the release will include unit tests covering exact matches, partial matches, malformed outputs, duplicated predictions, and empty outputs.

C.2 LLM-Judge Controls

LLM judges are used only when deterministic surface matching does not adequately capture semantic correctness, including translation and selected open-ended VQA tasks. The judge receives the task instruction, reference answer, and model prediction, and returns a constrained continuous score in $[0, 1]$. We use Gemini-3.5-Flash-Lite with deterministic decoding. The final release will record the exact model version, prompt, decoding parameters, retry policy, and evaluation date. Judge failures are logged and rerun under the same frozen configuration.

D Operational Routing Procedure

Before CGPO, we cache teacher competence and fit the calibration maps as described in Section 4.3. During policy optimization, each training sample follows this sequence:

1. Retrieve cached teacher competence $c_T(x)$, computed from K offline generations with the task-aligned verifier.
2. Generate G student rollouts, score them with the same reference and verifier, and compute $c_S(x)$ and $\Delta(x) = c_T(x) - c_S(x)$.
3. Evaluate the frozen calibration maps $\widehat{q}_T(x)$ and $\widehat{p}_{\text{win}}(x)$, then compute

$$\alpha(x) = \text{sg}\left[\widehat{q}_T(x)\,(2\widehat{p}_{\text{win}}(x) - 1)_+\right].$$

4. Compute the GRPO loss from the rollout rewards. For samples with a positive OPD weight, evaluate the frozen teacher on the student-generated prefixes and compute the length-averaged reverse-KL loss.
5. Update the student with $\mathcal{L}_{\mathrm{GRPO}}(x) + \lambda\alpha(x)\mathcal{L}_{\mathrm{OPD}}(x)$, keeping the teacher and calibration maps frozen.

When $\widehat{p}_{\mathrm{win}}(x) \leq 0.5$, $\alpha(x) = 0$ and the joint objective reduces to the GRPO term. Otherwise, the OPD contribution varies continuously with the calibrated teacher quality and superiority estimate.

E OCRBench v2.1 Evaluation Protocol

To comprehensively evaluate OCR capabilities, we adopt OCRBench v2 (Fu et al., 2025) as one of our primary evaluation benchmarks. OCRBench v2 is a large-scale bilingual, text-centric benchmark with 10,000 human-verified question-answer pairs spanning diverse visual scenarios. Throughout this report, subtask counts follow the evaluation export: 30 language-specific subtasks, comprising 21 English and 9 Chinese subtasks. These are aggregated into 13 language-specific task groups, comprising 8 English and 5 Chinese groups.

To further improve evaluation reliability and consistency, we conduct a systematic audit of both benchmark annotations and the official evaluation protocol. Our analysis identifies two aspects that may affect evaluation robustness: (i) annotation noise and format inconsistencies in reference answers, and (ii) evaluation criteria that may not fully align with the intended semantic objectives of certain OCR tasks.

Based on this analysis, we introduce **OCRBench v2.1**, a refined version of the benchmark that retains the original 10,000 samples and uses the same 30-subtask reporting scheme while incorporating manually verified annotation corrections and revised evaluation procedures. The refinement focuses on improving reference quality and metric consistency, without modifying the benchmark composition, task distribution, or evaluation scope.

All models reported in this work are re-evaluated under the OCRBench v2.1 protocol to ensure fair comparison. Detailed descriptions of annotation refinement and metric revisions are provided in Sections E.1 and E.2, respectively.

E.1 Annotation Cleaning

Motivation for annotation refinement. High-quality reference annotations are essential for reliable OCR evaluation, particularly for tasks involving structured prediction and semantic understanding. Under the original evaluation protocol, certain annotation issues may have limited impact because some metrics emphasize structural similarity or partial matching. However, under stricter content-aware evaluation, inaccurate references or inconsistent output formats can introduce additional penalties unrelated to the actual model capability.

In addition, ambiguous task instructions and inconsistent answer serialization may affect the stability of automated evaluation procedures, especially for structured outputs such as JSON, HTML, and field-value extraction. Therefore, improving annotation consistency is a necessary step toward more reliable and reproducible benchmark evaluation.

Scope of refinement. We conduct a comprehensive audit of all 10,000 samples, covering questions, answers, evaluation configurations, bounding boxes, and reference contents. The audit results in

Table 14 Annotation revisions in OCRBench v2.1. Type A normalizes answer serialization without changing semantics; Type B corrects genuinely wrong references; Type C rewrites extraction instructions to remove ambiguity. "#mod" counts revised samples per type; a sample modified in both `question` and `answer` is counted once in each type, so the column sums slightly exceed the 2,449 unique revised samples.

Type	Affected subtasks	#mod	Field
A. Format	chart parsing (en)	400	answer
	key information mapping (en)	300	answer
	table parsing (en / cn)	700	answer
	text grounding (en)	150	answer
B. Content	cognition / reasoning / diagram / science / math QA, APP agent, text translation (cn), KIE (en & cn)	626	answer
C. Question	key information extraction (en: 400; cn: 184)	584	question
Total	**16 of 30 subtasks**	**2,449**	

revisions to **2,449 samples (24.5%)**. No samples are added or removed, and the original indices, task categories, bounding boxes, and evaluation configurations remain unchanged. All modifications are restricted to the `question` and `answer` fields and are manually inspected for verification.

The annotation revisions can be categorized into three types, summarized in Table 14: (A) format normalization, which improves structural consistency without changing the underlying semantics; (B) reference correction, which fixes inaccurate or incomplete ground-truth answers; and (C) instruction refinement, which reduces ambiguity in task descriptions and output requirements.

Type A: format normalization (1,550 samples). We remove redundant formatting noise and standardize structured annotations to improve evaluation consistency without changing their semantic content. For table parsing, we eliminate scattered CSS styles, wrapper elements, and non-essential attributes, retaining only the basic HTML tags required to represent table structures. For other structured tasks, chart parsing and key-information-mapping references are converted from Python-dict-style representations into valid JSON strings, and text-grounding coordinates are uniformly serialized to ensure consistent parsing and matching during evaluation.

Type B: reference correction (626 samples). We correct inaccurate or incomplete reference annotations identified during manual inspection. These corrections mainly occur in multiple-choice reasoning tasks, key information extraction, and text translation.

Typical corrections include erroneous option labels in mathematical reasoning questions, swapped or inaccurate field values in KIE annotations, OCR-related text corrections, unified formatting of currency symbols, and revised translation references. For cases where multiple answers are semantically acceptable, all valid alternatives are explicitly enumerated and supported by the v2.1 evaluation protocol (see Section E.2).

Type C: instruction refinement (584 samples). We refine task instructions for key information extraction, where ambiguous extraction requirements may introduce uncertainty beyond the actual recognition and extraction capability.

For English KIE samples, we rewrite all 400 questions into a unified template that explicitly specifies the target fields to extract. For Chinese KIE samples, we append explicit JSON-format requirements and normalize field naming conventions for 184 questions. These refinements ensure that extraction metrics primarily reflect the model's ability to identify and organize information from visual inputs, rather than its interpretation of underspecified output formats.

Table 15 Metric revisions from OCRBench v2 to v2.1. Only the listed subtasks are revised; other subtasks follow the official evaluation protocol.

Subtask	OCRBench v2	OCRBench v2.1
Formula recognition	Substring matching (`gt in pred`)	Render-based CDM with token-level F_1
Document parsing	STEDS (structure-only similarity)	Content-aware md2md evaluation
Fine-grained OCR	BLEU, METEOR, token-F_1, and edit-distance average	Normalized edit distance
Full-page OCR	BLEU, METEOR, token-F_1, and edit-distance average	Normalized edit distance
Text translation	n-gram metric average	LLM-based semantic evaluation
Reasoning VQA	Single-answer exact matching	Multiple acceptable references

E.2 Metric Revisions

To improve the reliability and task alignment of evaluation, we revise the metrics for several OCRBench v2 subtasks where the original criteria may not fully reflect the intended capability being measured. Table 15 summarizes the differences between the original v2 protocol and the v2.1 evaluation scheme.

All remaining subtasks retain the original evaluation metrics, including exact matching / ANLS for VQA (Biten et al., 2019), TEDS for tables (Zhong et al., 2020), IoU-based metrics for grounding and spotting, F1 for information extraction, and normalized L1 for counting. The official aggregation strategy, including the 13 task groups and English/Chinese overall scores, is also preserved. Therefore, the reported differences arise only from the refined annotations and revised evaluation criteria described below.

(1) Formula recognition: from substring matching to render-then-match CDM. The original scorer determines correctness based on whether the reference formula appears as a substring of the prediction after removing whitespace. This criterion may be insufficient for formula understanding, as it can accept incomplete semantic matches while rejecting visually equivalent formulations with different LaTeX syntax.

We replace it with CDM (Wang et al., 2025a), where both reference and prediction formulas are rendered and compared at the visual token level. The metric performs token matching with geometric verification and computes an F_1 score:

$$F_1^{\mathrm{CDM}} = \frac{2PR}{P+R}, \qquad P = \frac{|M|}{|T_p|}, \quad R = \frac{|M|}{|T_g|}, \tag{14}$$

where T_p and T_g denote the visual tokens of the prediction and reference, and M represents the verified token matches.

This evaluation is more robust to common LaTeX syntax variations while remaining sensitive to the visual correctness of mathematical expressions.

(2) Document parsing: from skeleton similarity to content-aware md2md evaluation. The original implementation evaluates document parsing mainly through structural similarity. While this captures high-level organization, it provides limited assessment of whether the recovered textual content is correct.

We adopt the md2md protocol from OmniDocBench (Ouyang et al., 2025), where predicted and reference Markdown documents are decomposed into typed elements, including text, formulas, and

tables. Each element is evaluated using an appropriate metric, and the final score is computed as:

$$S_{\text{md2md}} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \text{Acc}(c), \qquad \text{Acc}(c) = \begin{cases} S_{\text{ED}} & c = \text{text}, \\ F_1^{\text{CDM}} & c = \text{formula}, \\ \text{TEDS} & c = \text{table}. \end{cases} \tag{15}$$

This makes document parsing scored at the character, formula-rendering, and table-structure levels simultaneously.

(3) Text recognition and full-page OCR: normalized edit distance. The original protocol combines multiple metrics, including BLEU, METEOR, token-level F_1, and edit distance. However, for OCR recognition tasks with relatively short outputs, these metrics may provide inconsistent signals and lack a unified interpretation.

We replace this aggregation with normalized edit distance:

$$S_{\text{ED}}(p, g) = \max\left(0,\ 1 - \frac{\text{EditDist}(p, g)}{\max(|p|,\ |g|)}\right), \tag{16}$$

which directly measures character-level recognition accuracy. The metric is deterministic, monotonic, and closely aligned with the objective of OCR text transcription.

(4) Text translation: LLM-based semantic evaluation. LLM-based assessment has been studied for both reference-based and reference-free translation evaluation (Kocmi and Federmann, 2023). Translation quality is primarily determined by semantic preservation rather than surface-level word overlap. Therefore, n-gram based metrics may not fully capture acceptable paraphrases, synonym substitutions, or different word orders.

We evaluate translation quality using an LLM-based judge, **Gemini-3.5-Flash-Lite**, with the fixed evaluation prompt shown in Figure 6. The judge evaluates semantic equivalence while considering translation completeness, factual consistency, and target-language correctness. The output score in $[0, 1]$ is directly used as the sample-level evaluation score. For samples with multiple acceptable references, the maximum score over references is adopted.

(5) Reasoning VQA (cn): multi-answer support. Some reasoning VQA samples may contain multiple semantically acceptable answers. The original evaluation protocol assumes a single reference answer, which may introduce unnecessary penalties for such cases.

The v2.1 scorer supports multiple valid references by accepting a prediction if it matches any enumerated answer. For single-answer samples, the evaluation behavior remains unchanged. This modification improves evaluation consistency for reasoning tasks with answer ambiguity.

E.3 Comparability and Reproducibility

The above refinements introduce two important considerations for evaluation interpretation.

First, OCRBench v2.1 scores should not be directly compared with the original OCRBench v2 leaderboard results, since the protocol incorporates annotation corrections and revised metrics for several subtasks. All OCRBench v2.1 results reported in this work are obtained under the same refined annotations and scoring procedures.

Second, the v2.1 evaluation pipeline is designed to be deterministic and reproducible. All metrics are computed offline with fixed procedures, with the exception of the LLM-based translation evaluator,

```
You are a translation evaluator for OCRBench v2. The task may be Chinese-to-English or English-to-Chinese. Compare the model's translated text with the reference translation and judge whether they express the same meaning in the intended target language.
Evaluation principles:
- Focus on semantic equivalence, not word-by-word matching.
- Natural paraphrases, synonym substitutions, and word-order differences should receive high scores if the meaning is preserved.
- Do not penalize harmless formatting or punctuation differences.
- Penalize mistranslation, missing key information, wrong entities/numbers/negation/ relations, or using the wrong target language.
- If the model output contains extra explanation, score based on whether the reference meaning is still complete and not contradicted.
Model translation:
{predict}
Reference translation:
{answer}
Please give a score between 0 and 1:
- 1.0: Semantically equivalent, fluent, and complete
- 0.7–0.9: Mostly equivalent with only minor issues
- 0.4–0.6: Partially correct but with obvious omissions or mistranslations
- 0.1–0.3: Only a few fragments are relevant; overall meaning is wrong
- 0.0: Completely wrong, unrelated, empty, or unreadable
Output ONLY a single number between 0 and 1 (e.g. 0.8). Do not output anything else.
```

Figure 6 The judge prompt used for the text-translation subtask in OCRBench v2.1. `{predict}` and `{answer}` are replaced by the model output and one reference translation, respectively.

which uses Gemini-3.5-Flash-Lite with a fixed evaluation prompt shown in Figure 6. The refined annotations and evaluation code are released together with this report to facilitate reproducibility.

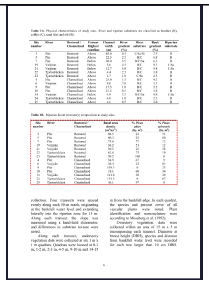
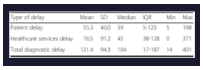
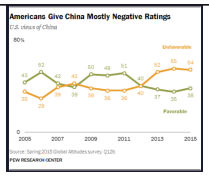
F Task Examples

Table 16 Data examples.

Type	Question	Answer	Image
Spotting	Spotting all the text in the image with line-level. Output the normalized coordinates of the left-top and right-bottom corners of the bounding box and the text content. The coordinates should be normalized ranging from 0 to 1000 by the image width and height. Your answer should be in the following format: [(x1, y1, x2, y2, text content), (x1, y1, x2, y2, text content)...]	[(148, 144, 907, 269, "北京北硬硬质合金有限公司"), (392, 234, 707, 334, "经营部"), (100, 606, 154, 621, "北河沿大街"), (96, 619, 161, 643, "11号")]	
Fine-grained OCR	Recognize the text within the [295, 390, 352, 432] of the image. The coordinates have been normalized ranging from 0 to 1000 by the image width and height.	679	
Text Recognition	what is written in the image?	JCPenney	

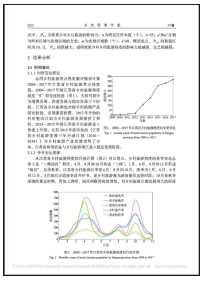
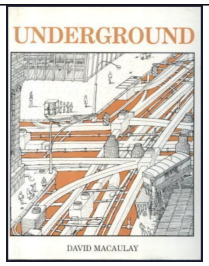
Continued on next page

Table 16 (continued)

Type	Question	Answer	Image
Text VQA	what are the letters on the tail section of the plane? Output the answer with 'answer' and 'bbox'. 'bbox' refers to the bounding box position of the 'answer' content in the image. The output format is "answer:gt, bbox:(x1,y1,x2,y2)", where the bbox is the coordinates of the top-left corner and the bottom-right corners. The 'bbox' should be normalized coordinates ranging from 0 to 1000 by the image width and height. Your answer should be in the JSON format: { "answer": "..", # The answer; "bbox": "(x1,y1,x2,y2)" # The bounding box position of the 'answer' }	{"answer": "EC", "bbox": "(240,530,295,576)"}	
Key Info Extraction	从图中提取: 样品特征和状态, 主检, 检验类别, 审核, 样品名称, 并按json格式返回	{"样品特征和状态": "外观正常", "主检": "###", "检验类别": "委托送样检验", "审核": "###", "样品名称": "玉米芯颗粒粉"}	
Formula Recognition	Please write out the expression of the formula in the image using LaTeX format.	$\mathbb{E}_{x \sim p_t}\left[\log \left(1 + \eta_t m(\lambda_0,x) \right) \right] = (1 - p) \log(1 - \eta_t \alpha) + p \log(1 + \eta_t (1-\alpha))$	
Table Parsing (region)	Describe the specific HTML-formatted structure of the table in the region [117, 419, 881, 656] of the image. The coordinates have been normalized ranging from 0 to 1000 by the image width and height.	<table> <tr> <td>Site number</td> <td>River</td> <td>Restored / Channelised</td> <td>Basal area density ($ m^{2} $ ha $ ^{-1} $)</td> <td>% Picea abies (by $ m^{2} $)</td> <td>% Pinus sylvestris (by $ m^{2} $)</td> </tr>.....</table>	
Table Parsing	Reconstruct the table in the presented picture as an HTML-style table in text.	<table> <tr> <td>Type of delay</td> <td>Mean</td> <td>SD</td> <td>Median</td> <td>IQR</td> <td>Min</td> <td>Max</td> </tr>.....</table>	
Chart Understanding	Is the value of Favorable 38 in 2015?	Yes	

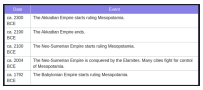
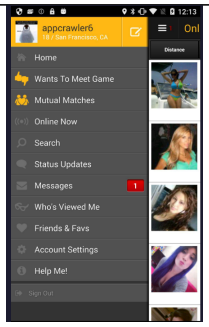
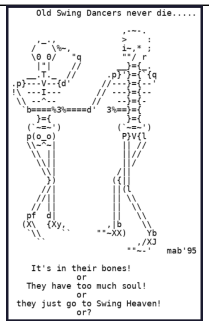
Continued on next page

Table 16 (continued)

Type	Question	Answer	Image
Full-page OCR	Read and output all the text content from this image.	第22卷第6期 2020年12月 辽宁工业大学学报(社会科学版) Journal of Liaoning University of Technology(Social Science Edition) Vol.22,No.6 Dec.2020 DOI: 10.15916/j.issn1674-327X.2020.06.009 区块链技术在企业财务舞弊治理中的应用 ..... 图1 虚增存货数总额 (C)1994-2020 China Academic Journal Electronic Publishing House. All rights reserved. http://www.cnki.net	
Document Parsing	*[System prompt: convert document page images to Markdown. Transcribe all visible text preserving reading order and structure; convert formulas to LaTeX; convert tables to HTML; do not describe figures; return only the converted Markdown.]*	2852 自然资源学报 35卷 式中： $ P_{D,U} $ 为要素D对乡村旅游的影响力；n 为研究区样本数（个），n=55； $ \sigma^{2} $ 和 $ \sigma_{i}^{2} $ 分别为所有区域与次级区域的方差；m 为次级区域数（个）。 $ \sigma^{2} \neq 0 $，模型成立。 $ P_{D,U} $ 的取值区间为 [0, 1]， $ P_{D,U} $ 的值越大，说明要素D对乡村旅游热度的影响力就越强，反之则越弱。 ..... 图2 2009—2017年江苏省乡村旅游热度的月统计图 Fig. 2 Monthly count of rural tourism popularity in Jiangsu province from 2009 to 2017 (C)1994-2020 China Academic Journal Electronic Publishing House. All rights reserved. http://www.cnki.net	
OCR-VQA	What is the title of this book?	Underground	

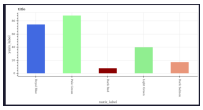
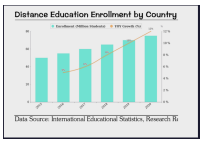
Continued on next page

Table 16 (continued)

Type	Question	Answer	Image
Education & Science QA	After the Akkadian Empire ended, many cities and empires tried to control Mesopotamia. Around the 1790s BCE, which empire started controlling Mesopotamia? A. the Elamite Empire B. the Neo-Sumerian Empire C. the Babylonian Empire D. the Akkadian Empire	C	
Scene Text Translation	What is the Chinese translation of the text inside the rectangle defined by [357, 137, 659, 309] in this image? The coordinates have been normalized ranging from 0 to 1000 by the image width and height. Please provide the translated content directly.	爱	
APP Agent	How many messages are unread?	There is 1 unread message.	
Text Counting	How many times does the character 'F' appear in the picture? Please output the exact number without any additional explanation.	2	
ASCII Art QA	What is depicted in this ASCII art image? A. dancing B. eating C. running D. singing	A. dancing	

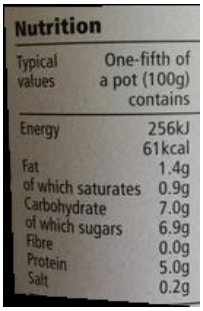
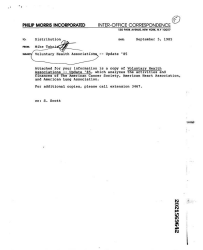
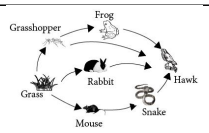
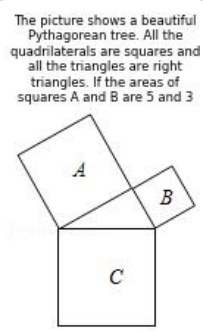
Continued on next page

Table 16 (continued)

Type	Question	Answer	Image
Chart Parsing	Convert the key information in the chart into a nested python dict.	{'title': 'title', 'source': 'Data source: ', 'x_title': 'xaxis_label', 'y_title': ['yaxis_label'], 'values': {'Royal Blue': {'xaxis_label': 'Royal Blue', 'yaxis_label': 74.5866897989292}, 'Pale Green': {'xaxis_label': 'Pale Green', 'yaxis_label': 88.21400944286917}, 'Dark Red': {'xaxis_label': 'Dark Red', 'yaxis_label': 7.675044699564133}, 'Light Green': {'xaxis_label': 'Light Green', 'yaxis_label': 39.7387773448238}, 'Dark Salmon': {'xaxis_label': 'Dark Salmon', 'yaxis_label': 17.0608316600648}}}	
Chart Parsing	Convert the key information in the chart into a nested python dict.	{'title': 'Distance Education Enrollment by Country', 'source': 'Data Source: International Educational Statistics, Research Reports', 'x_title': 'None', 'y_title': ['None', '%'], 'values': {'Enrollment (Million Students)': {'2015': '50', '2016': '55', '2017': '60', '2018': '65', '2019': '70', '2020': '75'}, 'YOY Growth (%)': {'2015': 'None', '2016': '5.0%', '2017': '6.0%', '2018': '8.0%', '2019': '10.0%', '2020': '12.0%'}}}	
Text Grounding	Where is the region of the text "HOLLYWOOD"? Output the normalized coordinates of the left-top and right-bottom corners of the bounding box. The coordinates should be normalized ranging from 0 to 1000 by the image width and height. Your answer should be in the following format: (x1, y1, x2, y2) # x1, y1, x2, y2 are the normalized coordinates of the bounding box.	(209, 292, 411, 335)	

Continued on next page

Table 16 (continued)

Type	Question	Answer	Image
Key Information Mapping	According to the information in the image, please pair the corresponding keys and values below: Keys that need to be paired are 'Total Fat per 100g/ml', 'Protein per 100g/ml'. Values that need to be paired are '5.0g', '1.4g'. Extract the required information in the text, and return the result in a dict directly. Your answer should be in the JSON format: {"key1": "..", "key2": "..", . . . }	{"Total Fat per 100g/ml": ["1.4g"], "Protein per 100g/ml": ["5.0g"]}	
Handwritten Answer Extraction	在单项选择题第4题中，考生答题的手写内容是什么？请输出考生选择的选项	B	
Document Classification	What type of document is the image? Here is the options: letter, form, email, handwritten, advertisement, scientific report, scientific publication, specification, file folder, news article, budget, invoice, presentation, questionnaire, resume, memo. Directly output the results without any additional explanation.	memo	
Diagram QA	What does the rabbit eat? Snake, Grass, Hawk, or Frog?	Grass	
Math QA	Please use the information from the provided image to directly answer the mathematical question described in the image.	8	
Reasoning VQA	what was the total number of points for alain prost and nelson piquet?	15	